

Radiolaria OS

Computing, Assembled Around Intent

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Computing, Assembled Around Intent

Practical AI applications must coordinate useful work while requirements, resources and permissions change. Radiolaria OS provides an execution kernel for this activity. Semantic participants propose task meaning; runtime materializes bounded Work; programs, services and numerical tools contribute typed results. Local owners retain decisions about acceptance and consequential actions. An integration supplies domain knowledge, capabilities and validation rather than granting a model unrestricted control.

The reference cases expose complementary relationships. Football connects a permitted schedule change to another owner's computed offer and a separately authorized mock booking. Wedding follows model-returned fields into actual computation, and historical quantum measurements through original-condition checks, consumption and separate saves. Ephemeral Workspace adapts media work after audio loss while retaining photo results, then completes its resource cleanup. Finite experience profiles change subsequent inspection or allocation without making advice a permission. External authorship demonstrates a separately examined domain implementation whose author did not acquire admission authority.

The contribution is a common organization of execution, continuation and extension, not a substitute for domain expertise, numerical methods or deployment safeguards. Source-linked records make each case and its limits inspectable. The evidence combines explicitly distinguished fresh and historical profiles, including controlled execution, captured provider responses and mock effects; it establishes neither universal security nor quantum advantage. The resulting application kernel gives integrators a concrete basis for applications in which intelligence contributes to work without owning its consequences.

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Equations: [Equation 1](#); [Equation 2](#); [Equation 3](#); [Equation 4](#); [Equation 5](#)

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01. Computing, Assembled Around Intent

A real task outlives an individual model response. Finding football pitches is useful only if the permitted dates, surfaces, budget, counterparty conditions and later consent still agree. Planning a wedding is useful only if the arrangement respects the organizer's current conditions. Reviewing photographs remains the same task when an optional audio participant disappears. In each case, the person wants an outcome, not the job of manually transferring every intermediate state between applications.

Radiolaria OS organizes supported computation around that continuing activity. Its execution kernel, RDL Kernel, combines semantic proposals, ordinary programs, bounded services and other computational resources through explicit input, result, time and ownership relationships. The public phrase "Computing, assembled around intent" describes this organization. It does not mean that a new physical computer is fabricated or that an unconstrained model invents any program. An integrator supplies the profile, available capabilities, typed contracts, adapters and policies. The kernel turns a permissible proposal into actual Work and records which results were used. [R-ARCH](#), [R-CONTRACT](#)

From Intent to Governed Activity

Work is a bounded unit of computation whose inputs, execution and result can be identified. Root is the local owner-controlled decision role, not a global supervisor over other owners. DRS denotes the retained-material discovery and retrieval mechanisms, whose contents inform decisions without preserving permission. BSEP, the Bounded Semantic Evidence Packet, is the semantic membrane through which side-specific task material is disclosed. A result records what computation produced; it is not permission to act. [R-ARCH](#), [R-CONTRACT](#)

The useful change is organizational. The person chooses the purpose, significant conditions and consent; the system carries the supported execution relationships through changing circumstances. Applications do not disappear. They can supply a renderer, solver, source, storage service or effect adapter. Only the capabilities needed for the admitted task participate. In EWS, the preview, controls and media services form a temporary environment; ending that environment need not delete an approved result or the history explaining it.

A short football example makes the distinction concrete. The original EXACT request does not yield a complete offer. Allowing only 8 January to move thirty minutes produces three natural-grass sessions for EUR63, within a EUR70 limit. That result is computed work, not a booking. It is wrapped with evidence and consumed by a different local owner. A later BOOK request and its own current decision produce one mock effect. REPEAT and VERIFY inspect the resulting history without another counted effect. The recorded episode explains each object and observation rather than treating the successful price as sufficient proof.

The account now returns to the whole construction. Wedding changes the objective and later two placement conditions, and Work consumes the changed meaning. EWS changes the available environment while preserving photo work. Numerical experience changes which inspection receives scarce Work units. External authorship changes who writes a domain extension while leaving independent examination and source admission in place. These variations are connected by the same distinctions between a proposal, material actually consumed, an applicable result, a local decision and an effect.

What OS Means Here

Radiolaria is an application execution kernel and reference implementation, not a bootable replacement for macOS or Linux. The accepted technical package contains finite profiles and reviewed reference code. It does not demonstrate autonomous discovery of arbitrary services, arbitrary-code confinement, industrial deployment or universal correctness. Those boundaries specify where the positive result is credible; they are not the product's sole purpose.

The broader design thesis is that capability growth should not entail authority growth. A better model can propose a better plan. A solver can compute a better candidate. An external author can supply a new

implementation. None acquires the authority to approve its own consequences merely by being useful. Intelligence and organization complement each other: improving one does not eliminate the need for the other.

This review corpus is bound to accepted technical commit A, not to a new runtime campaign. Its three deep episodes and cross-case comparisons explain recorded results and their exact scope. It offers four entries: understand the argument; verify selected saved evidence; build through the existing engineering interfaces; or read an inert, source-bound packet with an LLM. The publication itself is still an unpublished review proposal.

02. One Activity, Seven Views

An activity is not adequately described by a single sequence of boxes. Work may be nested, two owners may review different aspects independently, and retained material may outlive the process that produced it. The Geometry Atlas therefore asks seven questions about the same stable entities. It is not a new runtime architecture and it is separate from the Incident-to-Proof Atlas, which classifies finite boundary controls.

Composition Is Material, Not a Drawing

An intention enters a local Root intake boundary. Routing and semantic participants propose bounded structure; runtime materializes its own execution topology. The semantic membrane, BSEP, controls the meanings and side-specific projections exchanged. A topology is an execution structure, not an authorization. Work invokes actual capabilities with typed inputs and produces results and evidence. Downstream consumption must identify the field and result actually used; a report that merely names an upstream program is not enough.

Nested Work allows a parent to consume results from child cells. Retained execution can combine an unchanged historical child result with a recomputed selected child result in the prescribed parent-slot order. This supports local change without rewriting unaffected history. It does not establish a universal recursive planner. The experience chapter distinguishes the concrete retained-work and missing-capability mechanisms from the stronger, unexecuted picture of arbitrary child-local memory search.

Owners Are Not Stages in a Hierarchy

Root denotes a role/type: the sole local final and commit authority. VenueRoot and ActivePlanetRoot are separate instances with different sources and decisions. In the Airline profile, client, airline and bank remain independent. An airline offer, a bank authorization reference and a client's intent can inform a transaction without making one owner the superior business Root of the others.

Route acceptance, result acceptance and action commitment also differ. A model or orchestrator proposes; Post V&V checks; GT advises; Root decides. A consequential effect requires a current scoped ActionCommitPacket and the exclusive Effect Firewall/bounded Corridor. EvidenceReceipt records what happened. Neither the receipt nor a previous ACCEPT creates a reusable permission to repeat it. Source admission is a further, design-time boundary: reviewed code can be installed under a separately authorized repository operation, but that gives it no current runtime action permission. [R-ARCH](#)

The Atlas Questions

The [common overview](#) links activity, retained context and local decisions. Its locator opens the seven views below. MODEL is a participant, PROPOSAL its returned object, and BSEP the side-specific semantic membrane. TIME is separate from MEMORY; SOLVER is separate from the original-condition CHECKER. VISUAL_SEAL and CRYPTO_SEAL name different objects. These are editorial aliases, not new runtime types.

GEO-1 follows intention into proposal, nested Work, actual materialization and a changed configuration. GEO-2 distinguishes ownership, review, execution and source admission. GEO-3 follows birth, waiting, change, refusal, continuation, consumption and ending through separate lifecycles. GEO-4 follows meaning, address, owner-held material, provenance and local or external memory. GEO-5 places semantic models, code, solvers, devices and remote computation in their actual environments. GEO-6 shows how outcomes affect advisory choice and Work expenditure. GEO-7 separates external creation of an extension from its independent examination and later use.

Every diagram labels relationships: DATA is not PERMISSION; PROPOSAL is not EXECUTION; DOCUMENT_REFERENCE is not DERIVATION. ROOT remains the type, while VENUE and REQUESTER keep their concrete identities across views. Historical pilot A maps to part of GEO-1/GEO-2, B to a fragment of GEO-3/GEO-4, and C to the starting point of GEO-7. They do not substitute for the full Atlas.

These views explain why different capabilities can participate in one task. The kernel supplies common relationships; profiles supply domain-specific meanings. A renderer and a quantum sampler need not become interchangeable, and their raw outputs need not have the same schema. What can be shared is an explicit path from a bounded input to a checked result, actual consumption and a new local decision. The diagrams are conceptual unless an inset explicitly identifies a recorded profile; visual similarity is not evidence that an execution occurred.

Delegating Work Does Not Delegate Final Authority

A parent may invoke bounded child Work and consume its actual result. That is a useful execution relationship, not a transfer of final authority. The child remains constrained by its admitted contract, scope and budget. Creating another node does not enlarge those bounds or make the node a Root. One Root per child is neither required nor implied.

The same distinction holds across owners. If owner A permits B to use a particular result, B cannot thereby grant C A's access or effect rights. C needs its own lawful basis at the relevant boundary. Legitimate bounded delegation remains possible; an owner's reviewed scoped operation is different from inheriting the owner's position. No top-level diagram box is a SuperRoot. [R-ARCH](#)

Invariants, Carriers and Selected Witnesses

These eight families summarize accepted law, not a new universal test result. The rule/carrier/check columns are normative or explanatory; the last column selects recorded positive and adverse witnesses, not a fresh run or proof of every possible application. The [exact accepted lock](#), sections 3-7, is the common source; profile-specific keys resolve each witness.

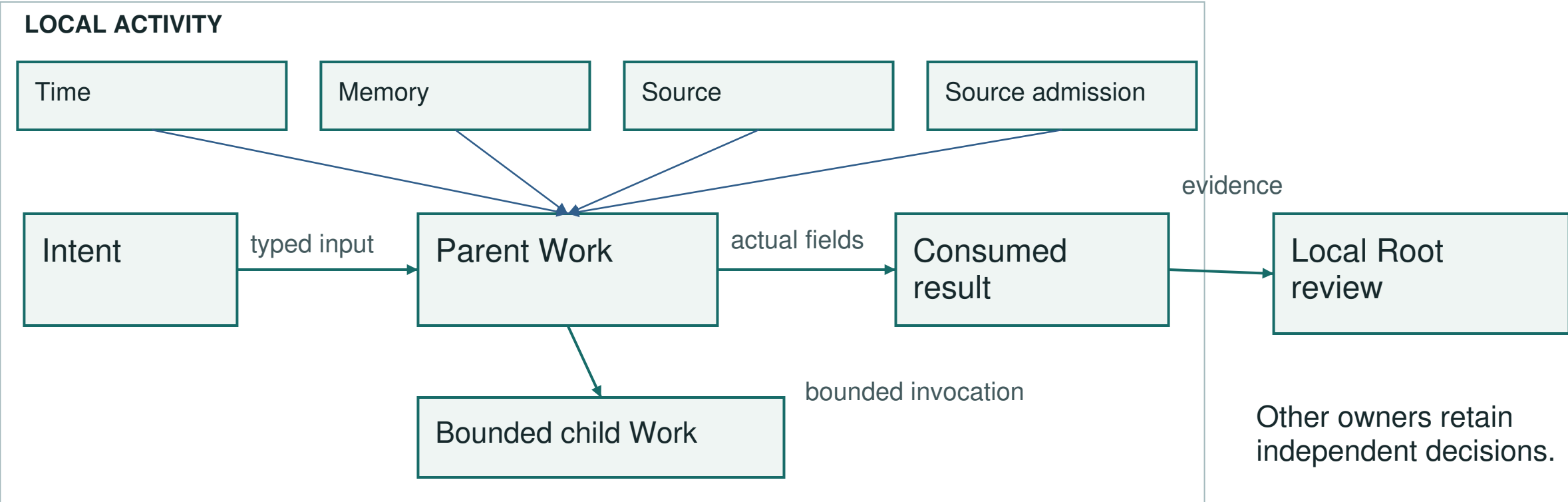
Rule and accepted section	Carrier and point of check	Selected lawful / adverse witness
Proposal is not authority, sections 3-4	BSEP and ResultProposal; local validation and Root review	Wedding accepted revision versus rejected incoherent proposal; W-W3 and W-C3
Topology is execution structure, section 4	RuntimeExecutionTopology; locally materialized inputs and exact consumption	Retained changed child consumed while unaffected history survives; X-TD-NAV (historical source route, not raw proof embedded here)
Child work stays bounded, section 4	Child contract, input, scope and budget; caller and runtime checks	Finite retained child result versus missing/mismatched dependency; X-TD-NAV (historical source route, not raw proof embedded here)
Owners decide independently, section 5	Owner-local source, decision and transaction	Venue offer consumed by requester; REVOKED source refused; F01-F04

Rule and accepted section	Carrier and point of check	Selected lawful / adverse witness
Effect needs exact current permission, section 6	Root packet, Host capture, Firewall, idempotency	Mock BOOK receipt versus refusal of stale/currently invalid material; F01-F04
Receipt does not authorize another action, section 6	Receipt and ledger remain historical evidence	EWS approved sidecar write versus stale packet refusal; WS-P_temporal, WS-P_save_packet, WS-P_save_receipt
Memory is not current permission, section 7	TemporalQuery, eligibility and fresh local review	G6A N4-04 cold/warm reuse: domain executions 1/0; current reads and Root checks remain. E08. Policy and nested-memory witnesses are separately indexed as E09 and E11.
Source admission is independent of runtime authority	Exact reviewed implementation identity and repository admission; no stored runtime grant	Authorized correction four followed by independent exam, while G54B1/G54D remain INCOMPLETE; R-AUTHOR, R-KIT

The source-admission row is a design-time boundary, documented by the lock's Repository Transition Review Boundary and the cited authoring sources. It is not derived from a business receipt. Specific evidence selectors and limits remain in the structured source key; a witness illustrates its named profile only.

One activity; several independent concerns

CONCEPTUAL MAP / Relations are not a recorded composite execution or a global owner



Time, memory and source qualify use. Source admission binds implementation. Current effects need their own packet.

03. History Remains; Current Use Must Qualify

Useful continuation depends on two apparently conflicting properties: history must remain stable, and present eligibility must change. A payment receipt should not be rewritten when a subscription expires. A photograph preview can remain correct when an unrelated audio process fails. A signed source can remain historically authentic after its current status becomes REVOKED. Radiolaria keeps these questions separate rather than forcing one mutable "valid" flag to represent all of them.

Several Clocks and Several Lifetimes

Time envelopes bind validity and observations to the relevant profile. Testflix's recorded Atlas control uses integer logical UTC seconds and strict evaluation before `valid_to`; an exact-deadline attempt refuses without executor work. Football also records fractional UTC observations of production, review, publication, receipt and consumption. Those observation timestamps are not its logical test clock or validity interval. Equal printed whole seconds do not prove ordering.

The accepted historical object is not a current Host. A captured response or serialized evidence file cannot reconstruct authority after a process dies. Public supplied validation can inspect historical meaning under its declared mode; a new action still needs current source/capture and Root conditions. This matters in a long-running activity: waiting, suspension, observation refresh and continuation do not silently extend a prior packet. In EWS, the old packet fails after audio loss even though a fresh changed-source observation later supports legitimate continuation.

Address, Material and Permission

A DRS address or descriptor points to material. It is not the material, and finding it is not permission to open or use it. The owner controls disclosure and descent. Local DRS can retain summaries and source-bound results; External DRS adds exchange across ownership boundaries. The receiving side must establish what was delivered, its source and applicable current status, then form its own consuming context and decision.

Football provides explicit objects rather than a metaphor. E is the producer Work's offer. B contains that value plus review/provenance and temporal material. C contains B plus the independently constructed currentness projection used by requester Work. The checked relationships are `B.offer=E` and `C.body=B`. They do not make E, B and C the same object or give their serialized forms equal bytes. The consuming field and Work invocation complete the evidence chain.

Memory Within the Task

Memory may be needed after the outer intake, inside a Work item or during continuation. The common composition implementation can observe a missing pure capability at a particular Work/input/current revision; local DRS discovery, current eligibility and Root-reviewed descent precede independent readmission and use. A stored program is therefore neither automatically installed nor automatically isolated. The bounded pure capability profile has explicit restrictions, not a promise about arbitrary Python.

Retained child results offer a different form of memory. Their historical fields and causal origin remain bound while selected descendants are recomputed. The parent must consume the correct canonical result order; validating the final result is not permission to substitute a different intermediate prefix. The technical chapter's source ledger records which claims are implemented mechanisms and which are test/checkpoint evidence. Arbitrary recursive child-memory activity remains a generalization, not an observed trace in these pages.

Ending Does Not Erase the Meaning

An environment can end while an approved sidecar, a receipt, or a summary remains inspectable. EWS separately requests saving, approves exact bytes and scope, writes a selection sidecar, then reaps owned processes and clears only its own unchanged cache files. A killed child receives no invented graceful-close receipt. Historical retrieval uses fresh review of the stored information; it does not recreate a closed workspace's authority.

The practical benefit is continuity with renewed grounds for use. Retained work reduces avoidable recomputation only when the current bindings still fit. Negative experience can recommend different scrutiny, not permanently revoke unrelated work. A digest proves a byte relationship under the chosen algorithm; a signature supports origin within its trust scheme. Neither establishes truth, current applicability, or the owner's permission by itself.

Source routes: [E09](#); [E11](#); [WS-P_memory](#).

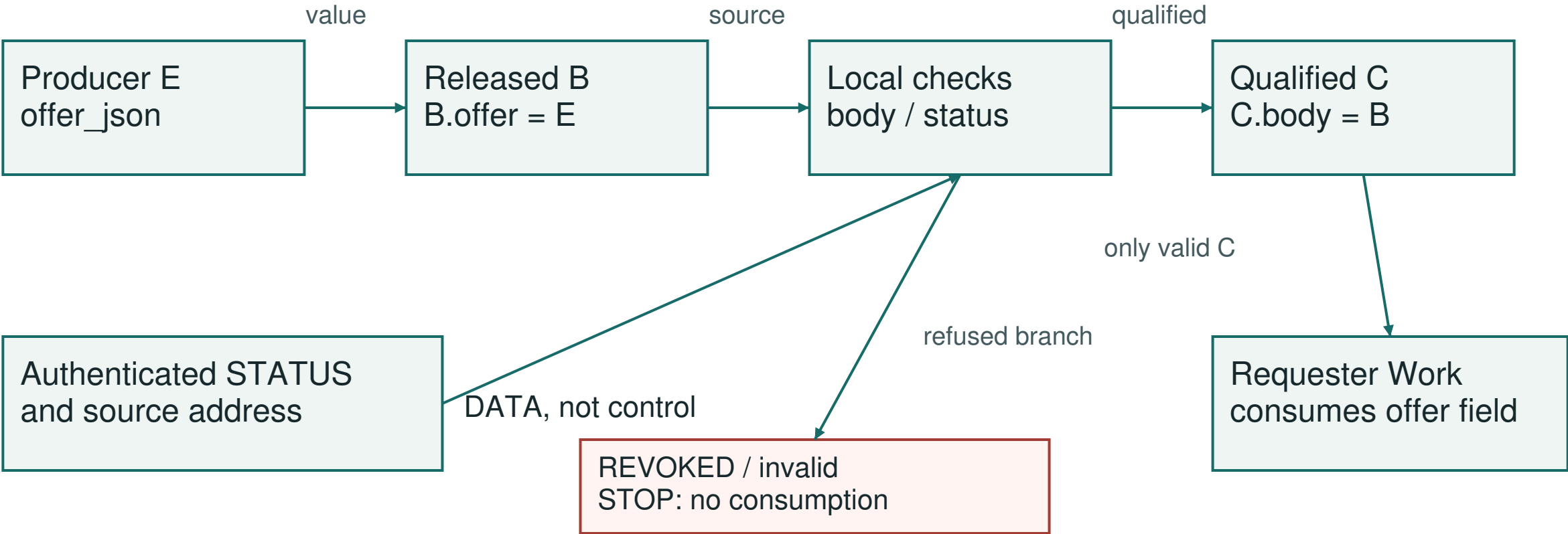
Accepted Definition Route

The accepted lock section 7 binds current retrieval to an explicit TemporalQuery and distinguishes immutable superseded history from current eligibility. Chapter 08's observed/received/validity fields are profile-local, not a universal wall clock. The exact general time definitions are indexed by R-CONTRACT in the structured source key; the bundled lock provides the common normative rule. A past result can remain informative while a current use requires fresh scope, policy and Root review. See [R-ARCH](#) and [R-CONTRACT](#).

The exact accepted [TemporalQuery schema](#) requires `as_of` and `freshness_bias`; its `historical_as_of`, `prefer_recent`, `strict_current` and policy-conditional mode remain distinct. [Common definitions](#), `/$defs/Timestamp` and `/$defs/TimeRange`, define date-time strings and nullable interval endpoints. These definitions constrain the representation, not a claim that every profile uses one universal clock.

Qualified context reaches Work; refusal stops

RECORDED FOOTBALL RELATIONS / E is a value inside B; B is material inside C



No arrow leads from refusal to Work. Booking still requires a separate current Root packet and action.

04. Different Participants, One Accountable Activity

The computer around an intention is not a single LLM. Semantic models can interpret a request or propose obligations. Ordinary code can transform media, evaluate a contract or calculate an exact rational. A solver can search a finite feasible set. A device or service can supply an observation or perform a bounded operation. A remote QPU can return measured candidates. Their competencies differ; the execution account must show which participant did which work and how its result became useful.

Meaning Can Participate During Execution

Wedding is a direct example. The orchestrator returns a task kind and needs; the requirement architect returns an objective and amendments. Those actual fields enter accepted local material and the compiler's input. KEEP_FAMILIAR and MIX_CIRCLES produce different seating arrangements under the same hard constraints. Later SR2 replaces two added placement conditions, and SR3 validates the existing result without another search. The model is not merely an external program author, and it does not select or authorize the hardware backend by implication.

The source origin must survive this explanation. Selected A and SR1 native episodes consume genuine prior captures through reexecution. B, VERIFY, AMBIGUOUS, SR2 and SR3 retain their separate LIVE_ROLE_ORIGIN records. Pure native $llm=0$ counters do not mean that the historical semantic acquisition had no provider calls. Conversely, reading those captures for this publication makes no new live call.

Numerical Candidates Need Original-Condition Checks

W4 uses a separately bound original problem and an explicitly chosen backend. The raw Rigetti sample, logical bit order, decoded seating, original-condition validation, selected result, native consumption and saved bytes have separate identities. A successful provider response includes invalid samples. Classical checks reject them; selection does not repair their seating. The selected valid sample matching a recorded optimum establishes a finite result, not quantum advantage or general device reliability.

The exact formalization belongs with its profile. Wedding defines binary guest/table indicators and nonnegative integer hard penalties; the objective expresses familiarity or mixing. G3/G4 uses fixed-point advisory scores and budget apportionment, not seating penalties. Calibration uses an exact reduced rational in abstract input units. One shared diagram must not imply one numerical law for all these computations.

Environments Also Differ

EWS materializes local services, a preview and controls for a finite task. Football's independent examination uses reviewed reference code and a declared container/runner boundary. Wedding's QPU is remote; the native consumer and save boundary remain local. These are distinct process, host and trust arrangements. A local mock effect is not a real purchase, and a local process check is not hardware remote attestation.

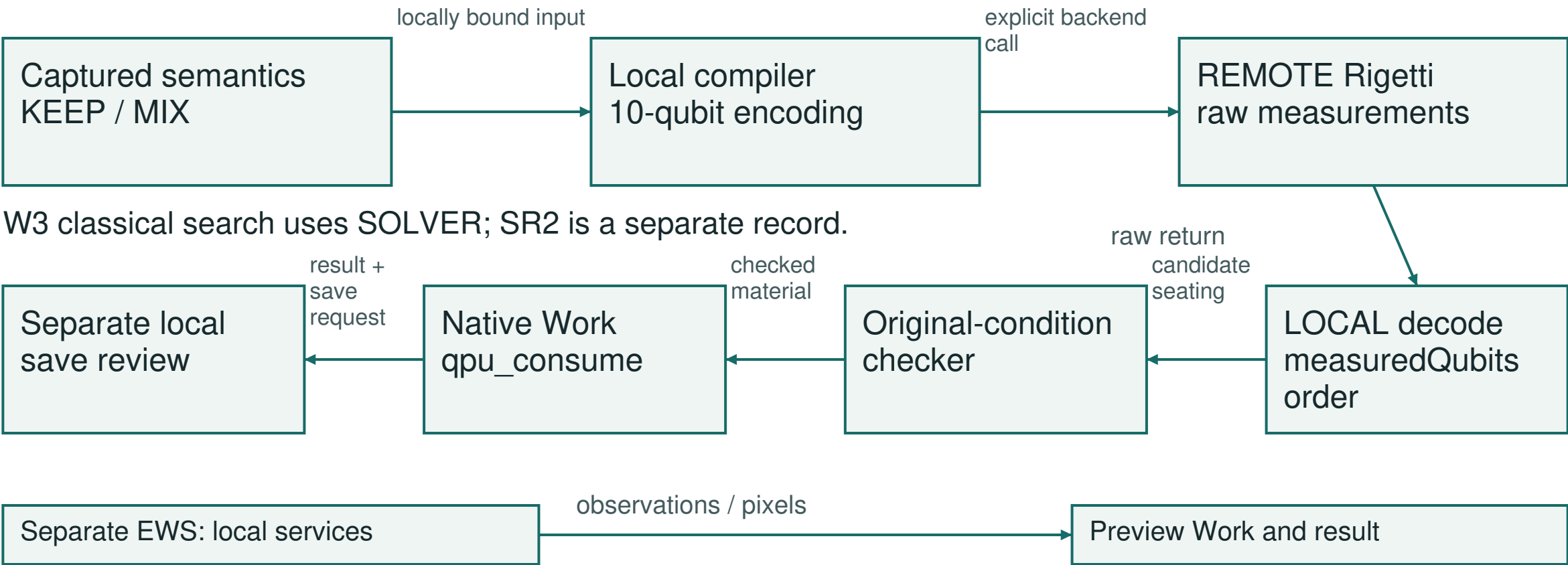
An integrator chooses and implements the supported adapters. IAM or policy engines can protect a service; a solver can remain a solver. Radiolaria supplies relationships for proposal, observation, Work, review, current action and history. It does not demonstrate automatic provider discovery or make all external dependencies safe. Optional semantic participation can be unavailable while a local profile continues useful work, as the Sentinel control illustrates; that finite result does not establish a real-time safety bound for every deployment.

The benefit is compositional rather than competitive: use the right computational participant for the admitted subproblem, then retain an inspectable account of its actual contribution. Better intelligence expands useful proposals only when the surrounding material, validation and decisions remain explicit.

Source routes: [W-W2](#); [W-Q2](#); [W-Q3](#); [R-CONTRACT](#).

Different computations share checked consumption

RECORDED W4 REVISION 1 / Captured W3 meaning is input, not permission for a QPU request



36-indicator QUBO differs from ten-qubit up-to-four-local encoding. The EWS inset is a separate local environment.

05. Experience, Choice and the Cost of Continuing Work

Radiolaria's useful memory is not merely a transcript. Within accepted finite profiles, an earlier result can change which review is proposed, where inspection work is spent, or whether an unchanged informational calculation needs repeating. The practical objective is to preserve useful work as activity continues without preserving yesterday's permission. Experience helps organise the next computation; the current owner still decides what may happen.

This chapter describes accepted source at 2e965ecb18e545e428380eb8e9aa5a7037a388be, using separately identified recorded executions. It does not claim a new run. The G6A evidence index distinguishes original, reviewed historical, and current retained-obligation sources; they are not one composite execution. Source and check references below resolve through the companion [ledger](#). [E01](#)

What an Outcome Can Teach

An Outcome Feedback Envelope distinguishes the quality of advice, the task outcome and enforcement. A bad proposal blocked correctly is negative evidence about that advice and positive evidence about the boundary, not a successful business action. Missing prospective expectations are not retrospectively manufactured. In the retained G35 five-domain report, fourteen unique occurrences yield zero scorable observations; those sources do not justify numerical learning merely because they contain completed work or refusals. [E02](#), [E04](#)

The G36R Supplier episode provides the more specific learning mechanism. Its ADV-3 record has UNSAFE advice, BLOCKED_AS_REQUIRED enforcement and SAFE_NO_DEAL, at CURRENTNESS, without a Firewall decision or effect receipt. Two earlier prepared-but-undispatched cases are NO_UPDATE. One eligible negative occurrence produces a sparse history, subsequently consumed through current local DRS/Root review. The selected review changes from standard to provenance; actual `supplier.review_provenance.v01` Work consumes the full bound material. A separate lawful continuation completes with one mock effect. The refusal therefore informs later scrutiny without becoming a permanent ban or a transferable authorisation. [E03](#)

That episode is historical captured reexecution of previously acquired provider-origin material. Its report's LIVE_OBSERVATION label must be read with the history lane CAPTURED_REEXECUTION; it is not a fresh provider call. Nor should it be joined to the separate controlled Airline experiment below as one task. [E03](#)

The numerical representation uses exact integers with scale $Q = 1,000,000,000$. A normalised value is its stored integer divided by Q ; these are dimensionless advisory quantities, not calibrated probabilities of safety. Signed rational rounding, RHE, means nearest integer with ties to even. For an eligible source-bound event, the implemented rating update is:

$$\text{Equation 1. } R_{\text{next}} = \text{clamp}(R + \text{RHE}(125000000 (O - E) / Q), 0, Q)$$

Exact implementation form

$$R_{\text{next}} = \text{clamp}(R + \text{RHE}(125000000 * (O - E) / Q), 0, Q)$$

Here R and prospective expectation E are integers in $[0, Q]$; observed result O is either 0 or Q . The expectation is fixed before the outcome, not substituted with the preceding rating. Comparable subject/profile/lane, non-future event time and the 64-effective-event limit are premises. Redelivery does not create another independent occurrence. Ineligible or unscorable events preserve the rating. `_evaluate_from_plain` implements the rule; source-bound validation and named rounding/deduplication checks are in [E02](#).

Inset 1: Negative Experience Reallocates Inspection

In the recorded G4 Airline comparison, a controlled prediction is incorrect even though enforcement is `ALLOWED_AS_REQUIRED`. The update takes $R=500000000$, $E=Q$, $O=0$ to 375000000 . The separately computed signed prior starts at zero and follows:

Equation 2. $\text{signal} = 2O - Q$

$$P_{\text{next}} = \text{clamp}(P + RHE(62500000 (\text{signal} - P) / Q), -Q, Q)$$

Exact implementation form

```
signal = 2*O - Q
P_next = clamp(P + RHE(62500000 * (signal - P) / Q), -Q, Q)
```

Thus one admitted negative occurrence gives $P_{\text{next}}=-62500000$. This is SPARSE, not a mature reputation. `fold_avf_history_prior_v01` uses the GT fold's admitted occurrence set and comparable history keys. [E02](#), [E05](#)

G4 uses that raw prior once in its five-feature pressure rule:

$$\text{Equation 3. } z_i = RHE((4r_i + 2l_i - 2u_i - c_i + p_i) / 8)$$

$$w_i = RHE(10^{12} \exp((z_i - \max(z)) / 250000000))$$

Exact implementation form

```
z_i = RHE((4*r_i + 2*l_i - 2*u_i - c_i + p_i) / 8)
w_i = RHE(10^12 * exp((z_i - max(z)) / 250000000))
```

The first four inputs are fixed-point integers in $[0, Q]$: configured relevance, lineage, uncertainty and estimated cost. The prior is in $[-Q, Q]$; weights are dimensionless integer apportionment weights. Hard-false branches are excluded first; unresolved required evidence prevents allocation, rather than being softened by the score. [E05](#)

Both offers have $r=l=800000000$, $u=500000000$, and lower/upper quotas of 2/6 dispatch units. Costs are 400000000 and 430000000. Cold pressures are 425000000 and 421250000; applying the negative prior only to offer 0 lowers its pressure to 417187500. The nine-unit original budget has already spent one unit on the initial constraint and reserves one final-consumer unit, leaving seven. Lower-first capped apportionment gives:

Offer	Cold weight	Cold units	Warm weight	Warm units
0	1000000000000	4	983881318977	3
1	985111939603	3	1000000000000	4

After allocating both minima, three units remain. Exact rational shares, floors and the largest remainder assign the last unit; stable identifiers resolve ties. Recorded performed-work lists confirm the allocation was consumed, not merely printed. Both instances use nine dispatch units. This is changed expenditure, not fewer total calls or a speedup. Implementation: `evaluate_reference_pressure_v01` and `allocate_reference_work_budget_v01`; checks and exact fractions: [E05](#).

The separate strategy contrast changes the configured utility profile from price-first to comfort-first and selects offer 0 versus offer 1. Its bounded feasible/individual-rationality/Pareto calculation is advisory. Client, airline and bank retain distinct decisions; the recorded hold is preparation, not an executed booking. [E07](#)

History Must Be Opened Again

Stored numerical history is not silently injected into the next task. The G4 retained evidence records descriptor discovery, Root approval, payload read, public-open validation and numerical source validation. The current bridge binds history to the receiving transaction. A later decision time requires requalification; expiry rejects current consumption while leaving the original evidence intact. [E06](#)

Age also changes advice. After separate time admission, `evaluate_gt_decay_value_v01` computes $T = RHE(R * 2^{(-a/h)})$, with integer age a and half-life h in seconds. Whole half-life multiples use exact rational division; the implementation returns zero at 64 half-lives. Only usable trust produces pressure $Q - T$; non-usable trust produces maximum pressure and recommends review. The implemented recommendation threshold is $T < 6000000000$. None of these calculations can suppress mandatory policy review or extend an expired advice window. E02 supplies the half-life premises, formula and checks.

Inset 2: Reuse Avoids Domain Work, Not Current Review

G6A's retained N4-04 witness addresses a different question: can the same informational answer be returned without recalculating it? Its native calibration input is integer readings $[8, 10, 9]$ and reference 10. The source formula is:

Equation 4. $s = \sum \text{readings} = 27; n = 3$

$\text{correction} = (n \text{ reference} - s) / n = (30 - 27) / 3 = 1 / 1$

Exact implementation form

```
s = sum(readings) = 27; n = len(readings) = 3
correction = (n*reference - s)/n = (30-27)/3 = 1/1
```

The output fields are `total=27, n=3, correction_num=1, correction_den=1`. Readings/reference are bounded integer domain values with no declared physical unit; correction is an exact reduced rational in the same abstract unit, and n is a count. `gate5_contracts_v01.calibration`, the native executor and `check_reuse` provide the implementation/check chain. [E08](#)

The cold observation counts one domain executor and one Host Work call. The warm observation counts zero of each, one stored read and two Root decision calls. Its answer must match the persisted summary, exact business inputs, owner, policy, schema and scope fingerprint. Changing a reading to $[8, 10, 8]$ produces `warm_business_binding`; expiry and substituting the cold Root decision also fail. Those observed function counts cover the current process/thread only; child activity is explicitly `UNOBSERVED_NOT_ZERO`. Setup and cold writeback are not inside the same measured window. No wall-clock speedup follows. [E08](#)

N4-05 makes the policy condition concrete. In a closed two-record store, validated replacement lineage and Root-reviewed writeback derive the current tip independently of list order. The predecessor remains historically valid, but a new current-policy Root review refuses its use. Summary-only descent opens zero payload bytes; restricted artifact descent returns `drs_pointer_access_policy_denied`. This is an explicit local policy composition, not a global supersession service. [E09](#)

Memory Inside Continuing Work

Memory need not occur only at outer intake. Implemented common-composition interfaces bind typed outputs to downstream inputs and retain exact historical fields across revisions. A missing pure capability is observed at a specific Work item, bound to its actual input, current task and revision. The bounded continuation path searches local DRS, evaluates current eligibility, obtains Root-reviewed artifact descent, retrieves source and independently readmits it before installation and use. Finding remembered code does not install it automatically. [E11](#)

Nested execution has a separate concrete basis: recursive review accounts for child cells, and retained-work checks consume one preserved child result with a newly computed selected child result in canonical parent-slot order. These are not evidence that every child autonomously searches memory. The stronger general picture, recursively created child-local DRS activity under arbitrary nesting, remains an architectural generalisation not demonstrated by the targeted sources. E11 distinguishes implementation, test assertions and historical checkpoint evidence.

The common outcome is useful continuity with renewed grounds for use. Scores remain advice; an independent Root decision and the exclusive effect path remain necessary for consequential action. The four approved programme deferments are N1-18, N1-19, N4-12 and N4-13. Full StrongGT/FullAVF, strategic regret/stability, robust/minimax/CVaR/sensitivity and LGT are not supplied by these bounded numerical profiles. External authorship does not cancel those deferments. [E10](#), [E12](#)

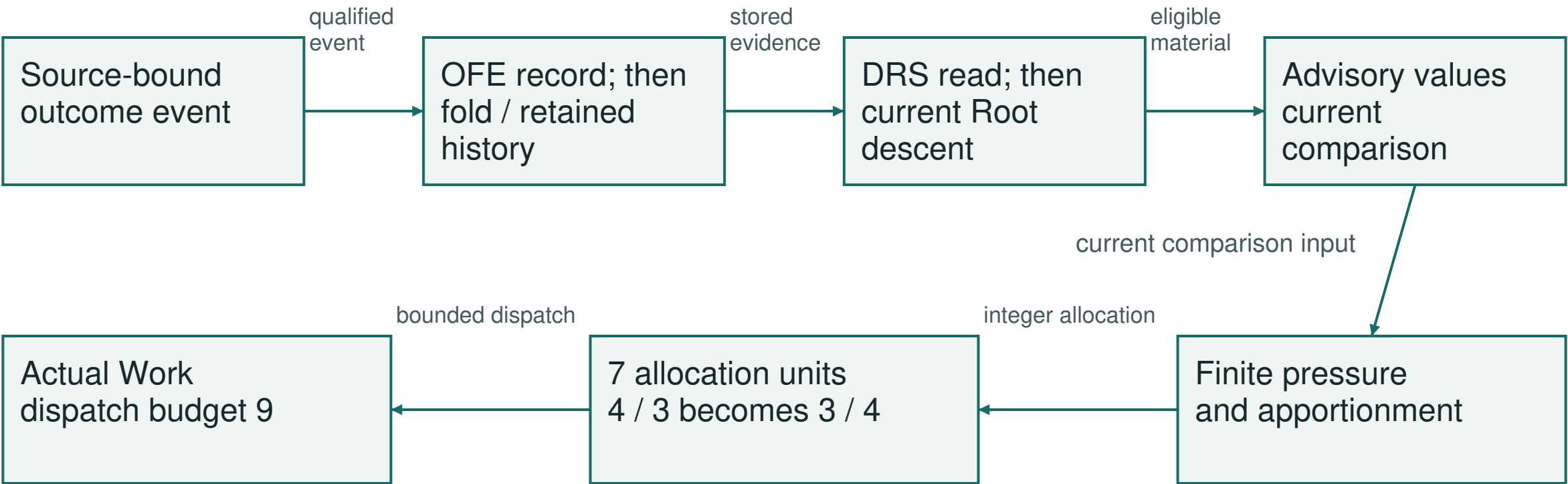
Inspect the Recorded Fields

[Source ledger](#).

Source routes: [E01](#).

Recorded experience changes finite expenditure

G3 / G4 RECORDED MECHANISMS / Eligibility and Root descent precede current advisory use



Budget 9 = 1 spent + 1 reserved + 7 allocated. N4-04 is a separate 1-to-0 domain-call witness with current reads and reviews.

06. What the Evidence Can Establish

The central evaluation question is not whether a model sounds persuasive. It is whether an intention or permitted change alters actual Work, whether the resulting material is consumed, and whether historical facts and current decisions remain distinguishable. A positive outcome is paired with a meaningful adverse or changed-condition neighbor. Useful continuation matters as much as refusal: a system that rejects everything would not demonstrate the intended organization of activity.

Five Evidence Classes

Accepted contracts define a finite profile's obligations. Recorded execution shows what particular code and inputs did. Independent checking can compare source-bound values or review an execution under stated trust assumptions. Architectural inference explains a broader relationship supported by several cases. Proposed future work is not execution. The editorial map records these classes rather than decorating every case with a green tick.

The current technical basis A is 2e965ecb18e545e428380eb8e9aa5a7037a388be, with tree 59a43cc0da470886804aab6c9af87fe17dfbd685. The accepted reference package reports 69 satisfied obligations out of 73 and four approved deferments: N1-18, N1-19, N4-12 and N4-13. Its campaign composition is TWO_FRESH_NINE_RETAINED. It would be false to describe eleven fresh profiles on one final source freeze. Historical source-impact bridges have their own scope; this editorial build does not refresh them. [R-FINITE](#)

Causal Evidence Is More Than a Successful Report

For a consumed result, the useful chain is actual input, producer result, wrapper/currentness material, downstream input and outcome. Football compares parent-held inputs and observed wire material with native Work; Wedding points to USED field references and retained output values; EWS connects the affected source and retained photo snapshot to the recomputed result. A candidate's self-report is not its own independent oracle.

The Incident-to-Proof Atlas makes a further distinction. DIRECT_CAUSAL cases connect the boundary event to later consumption. SHARED_DOMAIN_PROOF cases test another obligation on a common domain consumer. POST_CONSUMER_CONTROL cases happen after that consumer and cannot be its cause. Twenty cards are not twenty independent learners. Their source clocks, observations and native meanings must remain visible.

What Has Been Rechecked Here

This corpus performs data-only source selection, field comparison, link/format validation and editorial review. It does not rerun signatures, Football, Wedding hardware, the complete Gate6 verifier or any provider. Counts of inspected fields are publication QA, not new functional acceptance. Historical timing is reported with its measured scope: fewer domain executor calls does not by itself establish faster wall time, and nested inclusive timings cannot be added as independent cost shares.

The reproducibility route begins with inert saved objects. Pure supplied verification is a second route that consumes saved data and writes verification results. Fresh deterministic native execution is a third route with current inputs and dependencies. Live execution adds separately authorized network credentials, a provider/model and budget. These routes establish different things. A package manifest checked against itself proves consistency, not independent provenance; a reader needs a separately trusted commit or manifest digest.

Boundaries Are Testable Assumptions

Reviewed reference code, the observer and the Host/Firewall may share a process within the declared profile. That is a trust assumption to state, not a reason to pretend an unimplemented external action service exists. Conversely, an achieved bypass through an in-scope input remains a finding even if it reaches trusted code. The audit boundary cannot be moved after the fact to dismiss the result.

These finite observations do not certify production security, arbitrary-program isolation or universal intent handling. They do support an inspectable account of useful work across different tasks, participants, times and owners. The chapters preserve failures and lawful neighbors so that the reader can judge the actual construct rather than a curated sequence of success labels.

07. Football: An Offer Crosses Owners, Permission Does Not

A permitted change can make an offer useful without making it a booking. This episode follows a narrowly changed request into the venue's computed result, another owner's consuming Work and a separate mock-booking decision.

Classification: recorded G54D1 main p7_candidate_f779b790, newly cross-checked as data. Do not mix IDs with the later G6A football rerun. Negative neighbor: p7_candidate_7071661f, same G54D1 archive. Claim IDs CL02/CL03; Atlas B.

The useful change

EXACT returned NO_COMPLETE_OFFER. SHIFT revision 2 authorizes only January 8 to move by thirty minutes (consent:jan8-only). The independent parent input contains a cheap artificial field and field:elm, full-size natural grass at 2100 EUR minor units per session. Only elm meets the request. One lawful complete combination is independently enumerated from the retained input:

Session (Europe/Tirane)	Original	Selected	Price EUR minor
7 January 2027	16:30-18:00	16:30-18:00	2100
8 January 2027	11:00-12:30	11:30-13:00	2100
9 January 2027	11:00-12:30	11:00-12:30	2100

Total **6300 minor EUR = EUR 63**, under **7000 minor EUR = EUR 70**. The arithmetic does not make an unauthorized shift or surface acceptable. **F01**

Three distinct objects

E is parsed native producer outputs.offer_json. **B** wraps E in a source body with completed Work and review references, source revision 4 and a temporal envelope. B's raw file is 2890 bytes with SHA-256 1a9177b5b2748fa65193d0d6de3d268854b0924b773953d99bc136382d1279de. **C** is B plus a currentness projection, derived after the requester receives the bundle and status. The complete parsed requester inputs.offer_json equals independently reconstructed C.

E is not B, and B is not C. Independent construction of C uses the parent-held request/inventory, the actual received wire bundle and STATUS, not the candidate's claimed C as its own oracle. The data-only checker verifies Work IDs, source-review ID, wire hash and observed order. This proves the claimed data joins under the retained observer trust assumption; it does not newly authenticate signatures or establish remote attestation.

Producer Work:

work_results:b61c20eee7eb440fd73159a122a7e62405f699be41d7d7130489a1aaa876b6a9.
 Source review: 51c5d58f026068bacc1282a78c955ca97b5c740e165d82180dc072e827a68f81.
 Source record: d8eafc63c564cd4bfa0a0a40c4e0ea0f8832d153a0fc3271c7d29d518b80c808.

The observer's fractional UTC sequence places producer completion before source review, publication wire transfer, requester reception and consumer return. Artifact timestamps share integer second 1790359281; do not infer strict numeric timestamp growth from the envelope alone.

Address, material, release, currentness, permission

The pointer locates an object. B is the material. A signed release binds its body hash, recipient, request, nonce and deadline. STATUS says whether the source is currently usable; in this trace ACTIVE was checked at 1790359281, valid until 1790359341. The body is valid until 1790359581 with TTL 300. At use time 1790359281 the bounded windows overlap. Shape, integrity, provenance, scope, local policy and anti-rollback checks still matter; TTL is not permission.

ActivePlanetRoot performs its own import/result review and Work. VenueRoot's source decision is evidence, not a permission token for that other Root. SHIFT ends OFFER_READY, booking=null, receipt_ref=null, with zero dispatch and zero mutations. There is no SuperRoot coordinating a cross-owner authorization. [F02](#)

Negative neighbor

The recorded signed status is REVOKED. Application reason: status_not_active. The actual native Root decision is BLOCKED_FAIL_CLOSED, reason hard_identity_violation. This branch has no successful requester PURE consumer or effect. The historical signature envelope is preserved, but this editorial run does not redo cryptographic verification. A signature on a revocation cannot turn it into usable offer evidence. [F03](#)

BOOK is a new permission boundary

Only a separate CONFIRM MOCK_BOOKING request with owner_approval_ref=consent:controlled:booking enters the booking action. VenueRoot owns its registry and action decision. The retained observations count one native dispatch and one mock executor effect for a single three-slot batch, followed by source readback.

Receipt: effect_receipt_v01:37c2a6d339f25e63aad83fb378bbe6d815c281ad1e4e5ece034b3d745418c0a4. It is EvidenceReceipt owned by VenueRoot, not a live venue purchase and not transferable authority. REPEAT and VERIFY do not add another effect; history and final readback retain the same booking. Their saved success cannot authorize a new action today. [F04](#)

Inspect without executing

[Derived facts](#) retain all values used in the explanation; [primary source inventory](#) gives exact archive member paths, lengths and SHA-256. Full primary JSON remains local under references/episode/. [M2](#) embeds a narrower self-contained selection and code excerpts. G54D1's historical independent reviewer result remains historical; the author neither ran that recovered helper nor treated a candidate report as the independent oracle.

08. Wedding: Changing the Plan Without Losing the Conditions

The Practical Question

A seating plan is not finished merely because every guest has a table. The organizer may want familiar groups together, prefer people from different circles to meet, or change a particular placement after seeing a first result. Some conditions remain mandatory throughout. Can semantic interpretation, exact computation and an external numerical participant contribute to that evolving activity without silently changing what the organizer asked for?

The recorded Wedding profile makes this question inspectable. Twelve synthetic guests occupy three tables of four. Six pairs must sit together; guests 01 and 05 must sit apart; guest 11 must sit at table 3. Twelve familiarity pairs define the preference to optimize. These are deliberately small, explicit inputs, not observations about a real wedding. The useful outcome is a complete, checked seating arrangement, with a supported route for changing requirements and a separate route for saving a numerical result. [W-W1](#)

There are two related but distinct evidence stages. W3 connects human-language requests to model-returned semantic fields and native Work. W4 connects previously captured meaning to an explicitly chosen quantum backend, measured candidates, classical checks, native consumption and saved files. Neither stage is a story about an external model merely writing application code. Nor should they be edited into one uninterrupted execution.

A Model Response Becomes Work

Request A asks for a complete plan that keeps familiar circles together while retaining every mandatory restriction. Its two Gemini captures return different contributions: WEDDING_INTENT_ORCHESTRATOR returns `task_kind: GENERATE`, review/search/validation needs and requested outputs; WEDDING_REQUIREMENT_ARCHITECT returns `objective_profile: KEEP_FAMILIAR_V01`, the required capabilities and no amendments. The actual request body exposes permitted task and objective definitions, not a prescribed seating answer. [W-W1](#), [W-W2](#)

The returned fields do not acquire authority by being fluent or well-formed. `validate_capture_v01` checks request, role, source snapshot, response hash and serialized disclosure. `parse_payload_v01` checks the finite task, capability and amendment vocabulary. `accepted_material_v01` constructs the local owner context and material; its name describes accepted domain input, not a replacement for Root review. `_execute_material_v01` then builds the proposal, bounded Work and local review path. [W-C1](#), [W-C2](#)

For A, the recorded program orders review, compile, solve, validate, consume. Its compiler invocation really receives material containing the selected profile. The following Work records identify the producer result, downstream Work instance and `/material` field as USED, with `decision_effect: bind_input`. The consumed result is `[0,0,0,0,1,1,1,1,2,2,2,2]`, where 0/1/2 mean tables 1/2/3. A local final-result review records ACCEPT; it does not request an effect or create permission. This is evidence of use beyond the provider response alone. [W-W3](#)

Request B changes the preference to mixing circles while retaining the same original problem. Its architect actually returns MIX_CIRCLES_V01. Compiler input, validation and output retain that value. The result becomes `[0,0,1,1,1,1,2,2,0,0,2,2]`. The guest set, capacity rules and mandatory conditions stay fixed; the objective and seating change. A and B have separate request identities, `request:w3:v02:a` and `request:w3:v02:b`, rather than being two states of an invented common Host. [W-W3](#)

Origin labels matter. The selected A and SR1 native episodes are CAPTURED_PROVIDER_RESPONSE_REEXECUTION: genuine earlier role responses were consumed again,

without another model call. B, VERIFY, AMBIGUOUS, SR2 and SR3 retain LIVE_ROLE_ORIGIN. Their captures are historical live-origin records, not calls made for this chapter. Native `l1m: 0` counters cover the pure native lane, not the external semantic exchanges. SR1's original attempt-015 receipt remains FAILED, reason `collection_bound`; later captured consumption must not erase that parser failure. [W-W2](#), [W-W3](#)

Change Two Placements, Preserve the Rest

The SR sequence gives a more demanding example than changing a preference label. SR1 asks for mixing while putting guests 07 and 08 at table 2. Its returned group amendment becomes two typed `ALLOWED_TABLES` conditions. The accepted problem advances to revision 2, and the output places both guests at table 2. [W-W4](#)

SR2 asks: "seat guest_07 and guest_08 at table_1 instead of table_2." The architect's actual response in attempt 017 returns two `REPLACE` amendments. Each names its real prior added condition, its guest, tables: `["table_1"]` and source_ref: `request:w3:v02:sr2`. The model is not allowed to replace the eight protected base conditions. The parser also requires complete replacement of the prior added requirement, preventing old and new placements from accumulating inconsistently. [W-W2](#), [W-C1](#)

This change has an explicit acceptance and consumption chain. The material records a `SCRIPTED_CONTROLLED_OWNER_INPUT` acknowledgement with `browser_click_claimed: false`. The proposal payload and first Work input also carry a revision Root `ACCEPT`, decision `82c4e166cb503b4f8b5717c48ac6412be0d7d653614ee4e59908fc063fdc3627`. The revised problem is revision 3, with content reference `dd8c7848b55b1325bf75589eb710ce5e6d6fe0dee6e5fa36ec3dc99aa09aefdf`. This is scripted owner input plus recorded local review, not an observed interactive browser consent. [W-W4](#)

The compiler consumes the review result identified as `capability_result:17ade9ca7c6ba2079dd55f2a55c8acced426253d430cc60a46aab596c06d34d0`; its actual input contains the table-1 conditions. Solve, original validation and consumption follow with their own causal field references. The selected arrangement changes from SR1's `[0,0,1,1,2,2,1,1,0,0,2,2]` to `[0,0,1,1,2,2,0,0,1,1,2,2]`: guests 07/08 move to table 1 and guests 09/10 move to table 2. The eight base conditions and mixing objective remain. [W-W4](#)

SR3 asks only to check the current plan. Its request explicitly forbids rearrangement or alternatives; its architect returns a null objective and only `VALIDATE_ORIGINAL`. The accepted context inherits the objective from the actual prior result, rather than pretending that the model selected a new one. Its program contains only `validate` and `consume`, preserving the SR2 arrangement. SR1, SR2 and SR3 remain distinct native tasks, connected where their recorded prior-result references say they are connected. [W-W4](#)

The adverse neighbor is equally concrete. The old SR1 arrangement is `VALID` under revision 2 but `INVALID` under revision 3: the two new placement conditions fail, although its mixing score is still zero. `old_under_new.json` preserves both verdicts and `original_sr1_unchanged: true`. Its unfortunately named `current_root_refusal` field contains `current_owner_context`. In `NativeRunV01.validate_current`, that is the domain owner-context check before further validation, not a newly executed Root rejection. Separately, `AMBIGUOUS` returns `NEEDS_CLARIFICATION` with zero search calls instead of choosing an objective for the organizer. [W-W5](#), [W-C2](#)

What the Mathematics Checks

Let $x[g, t]$ be a binary indicator for guest g at table t . The original representation has 36 indicators. The nonnegative integer hard penalty H adds squared one-seat and four-guests-per-table deviations, pair-equality penalties for `TOGETHER`, same-table products for `APART`, and indicators for forbidden placements. $H = 0$ means all original conditions hold in this finite profile. [W-C3](#)

For the familiarity-pair set F , the two dimensionless soft costs are:

$$\text{Equation 5. } \text{KEEP}(x) = \sum_{(g,h) \in F} \sum_t (x_{g,t} - x_{h,t})^2$$

$$\text{MIX}(x) = \sum_{(g,h) \in F} \sum_t x_{g,t} x_{h,t}$$

$$Q(x) = P H(x) + \text{selected_soft_cost}(x)$$

Exact implementation form

```
KEEP(x) = sum_(g,h in F) sum_t (x[g,t] - x[h,t])^2
MIX(x)  = sum_(g,h in F) sum_t x[g,t] * x[h,t]
Q(x)    = P * H(x) + selected_soft_cost(x)
```

On feasible assignments KEEP counts each separated familiar pair twice; MIX counts each co-seated familiar pair once. With twelve familiarity pairs, the compiler chooses $P = 25$ for KEEP and $P = 13$ for MIX, one above the corresponding feasible soft bound. Given a feasible state, nonnegative integer H makes an invalid state's energy exceed every feasible soft score. This separation argument is not a claim that hardware finds the optimum. It is implemented by `ProblemMathV01.components`, `penalty` and `compile_optimization_v01`. [W-C3](#)

`solve_profiles` enumerates balanced assignments; SR2 records 34,650 evaluated assignments. `validate_assignment_v01` independently reports original-condition verdicts and objective components. The separate historical enumeration record gives ten feasible arrangements and five MIX optima for each of SR1 and SR2, with minimum zero. That recorded check is pinned prior evidence, not a new enumeration executed for this chapter. [W-W4](#), [W-W6](#)

A Measurement Is Still a Candidate

W4 deliberately returns to the original revision-1 problem, not the revised SR2 problem. `preparation_material_v01` consumes the captured A/B meaning under new W4 requests and binds a recorded local angle grid. Its backend field says `CURRENT_W4_OWNER_INTAKE`; historical model needs are explicitly not QPU permission. The numerical program bridge, runtime semantic roles, classical solver and QPU therefore have different jobs. Engineering supplied the supported integration; these records do not demonstrate autonomous provider discovery. [W-Q1](#), [W-C4](#)

The numerical encoding uses five variable guest pairs, two bits each, while the sixth pair is fixed at table 3. It is `PAIR_BINARY_10Q_V02`, an exact substitution with up-to-four-local terms, not a ten-variable QUBO. With measured logical-qubit columns `q[j]` and bits `z[j]`, `measurement_index_v02` computes $\sum_j z[j] * 2^{q[j]}$. Each pair code 0/1/2 maps to a table; code 3 is unseated and never repaired. [W-C5](#)

The KEEP provider result identifies Rigetti Cepheus - 1 - 108Q, task `2c896911-1bd9-4c72-9d03-873fb782b59b`, and 1,000 measurements. Raw row 63 is `[1,0,1,0,0,0,0,0,0,1]`, yielding basis 517 and arrangement `[1,1,1,1,0,0,0,0,2,2,2,2]`. Classical checking finds all eleven original verdicts satisfied and objective zero. Of the 1,000 samples, 33 are valid and 967 invalid; 16 distinct valid plans appear. `samples_v01` selects by objective, then assignment, then shot, with `local_repair: false`. The result matches the recorded local optimum; it establishes no quantum advantage. [W-Q2](#)

A measured adverse neighbor prevents a success-only account. KEEP shot 2 yields basis 128 and `[0,0,0,0,0,0,2,2,0,0,2,2]`. Its original-condition report fails capacities at tables 1 and 2 and the guest-01/guest-05 separation. The provider returned it successfully; the application cannot use it as a valid plan. [W-Q2](#)

The separate MIX task `470b472c-590e-42ad-b4f1-6b597be40f6d` returns 42 valid and 958 invalid samples. It selects shot 31, basis 292, objective zero. Its arrangement happens to equal SR2's arrangement, but its problem reference is the original revision-1 reference. Equal seating arrays are not equal problem, task or permission identities. [W-Q2](#)

From Checked Sample to Saved Result

W4's consuming native program is `qpu_validate`, then `qpu_consume`. KEEP's consumption record explicitly binds the first result's `/material` into the second Work instance as `USED`. The output retains raw SHA-256 `746f6671f68dff12c31f9637c9ddefc4fa1f1692868d569316275abb144bad0e`, shot 63, basis 517 and `LIVE_QPU_RAW_DECODED_POSTSELECTED`. It is not a locally repaired plan substituted for a measurement. Native result acceptance and current save permission are separate. [W-Q3](#)

`save_current_v01` checks the native run and submits exact bytes, path and a time-bounded local scope for a separate Root review. The KEEP JSON save receipt records `SAVED_READBACK_VERIFIED`, 520 bytes and SHA-256 `3756378338308c7f97d438519fabb3a64ae98b3c861b96e10aa46976666689c5`; the HTML has its own receipt. These are saved task-owned reports, not a reservation, invitation or physical seating action. [W-Q4](#), [W-C6](#)

The supported lesson is practical: interpretation can change what is computed, and a later request can validate rather than solve again, while original restrictions remain inspectable. External numerical candidates can join that activity without replacing classical checks or the owner's decisions. This chapter establishes no fresh runtime result, provider attestation, universal workflow synthesis or hardware speedup. Full identities, original-byte pins and exact field projections accompany the text; they make the account inspectable without turning recorded evidence into current authority.

Paper Citation Namespace

W-W1 through W-W6 identify the W3 intake, semantic and native records; W-Q1 through W-Q4 identify the distinct W4 numerical and save chain; W-C1 through W-C6 identify implementation sources, not new execution evidence. Full repository paths, historical identities, selectors and source classes are collected in the [paper source key](#). These editorial labels do not rename primary evidence.

Inspect the Recorded Fields

[CODE_AND_EDITORIAL_PROJECTIONS.json](#). [SOURCE_LEDGER.json](#). [CHAIN_LEDGER.json](#). [PROJECTION_IDENTITIES.json](#). [W4_PROJECTIONS.json](#). [CAPTURE_PROJECTIONS.json](#). [W3_PROJECTIONS.json](#).

Projection note: selected save-claim path strings are `REDACTED_DERIVED`. Their original-value digest and published-value digest are distinct; other selected fields remain exact. [Current projection identities](#). Original evidence is unchanged.

09. The Workspace Can Change While the Task Continues

A Task, Not a Permanent Application

A person wants to review photographs and a short video, keep useful selections, and then finish. The environment should bring the needed capabilities together without making the person reconstruct the task whenever a participant disappears. Ephemeral Workspace gives this idea a bounded, concrete form: local media access, preview rendering, display, controls, optional audio, and a separately gated save operation become participants in one temporary activity.

The question is not simply whether a tailored interface can appear. Can the environment lose a capability, continue useful work without discarding unaffected results, save only what was separately approved, and end while leaving inspectable evidence? The principal recorded EWS3R2 browser run answers that question for synthetic local photo/media work. A separate historical EWS3 profile demonstrates later informational retrieval. These are related demonstrations, not one invented continuous session. This chapter is an editorial

account bound to accepted source HEAD 2e965ecb18e545e428380eb8e9aa5a7037a388be, not a new execution. Sources: [WS-appendix, WS-claims, WS-index in the source ledger.]

Birth: Available Capabilities Become Actual Work

The mixed request asks to browse, rate and select photographs, adjust exposure and crop, and review video with audio. It explicitly permits silent visual continuation if audio disappears, forbids modification of originals and publication, and requires separate confirmation before saving. The fixtures comprise six synthetic photographs and an eight-second, sixteen-frame sequence, with mono 8 kHz PCM tones. They are not camera RAW development, compressed-video decoding, recorded speech, or proof of sound from a physical speaker.

Three semantic contributions make the task precise. The orchestrator supplies needs; the architect supplies preview, cleanup, save and audio obligations; the privacy reviewer supplies minimization and an allow/block decision. In the principal run these were three CONTROLLED_DETERMINISTIC outputs from local-semantic-rules-v01, with no provider call. In particular, audio_policy="SILENT_CONTINUE" and minimize="preview_without_metadata" affect subsequent work, rather than merely decorating a report.

Runtime materializes compile_contract and consume_contract: the second Work item consumes the first item's actual output. Root route review and later command authority remain separate from semantic proposals. The finite contract organizes scoped local services; it does not discover arbitrary devices or let a model author an unrestricted execution graph. One persistent Host and local Root retain the action boundary. Exact selections: [WS-P_controlled_0 through WS-P_controlled_2, WS-P_topology.]

What the Photo Controls Actually Mean

The interface supports manual rating and selection. RATE stores a chosen integer from zero to five; it is not an aesthetic model score. The principal scripted browser sequence selects asset : 1, sets rating five, exposure five tenths and SQUARE crop. The interaction is real browser activity, but SCRIPTED_BROWSER_TEST is not evidence that the owner personally clicked. Even an endpoint labelled MANUAL_OWNER_CONFIRMATION does not change that provenance.

Exposure is a preview brightness transform, implemented as $2^{**}(\text{exposure}/10)$, where exposure is an integer in tenths. It is not a camera sensor or RAW-pipeline claim. SQUARE and WIDE select centered aspect-ratio crops; ORIGINAL leaves the aspect ratio unchanged. The renderer converts input to RGB, applies these transforms, resizes within the preview bounds, and emits a fresh PNG. Originals remain unchanged; the preview is a derived representation.

Under the principal privacy policy, display receives derived pixels and bounded frame information, not source paths or embedded metadata. Removing metadata does not hide visible image content, erase pixels already viewed, or isolate processes sharing the same host account. These distinctions explain both the useful feature and its actual boundary. Sources: [WS-render_code, WS-session_code; selections WS-P_sidecar, WS-P_complete.]

Change: Audio Disappears, the Task Does Not

After four delivered video frames and 16,000 PCM samples, the demonstration deliberately kills its owned AudioSink. This is an actual process loss, not a checkbox simulating unavailability. Media work pauses pending a fresh observation and continuation decision. Photo work remains retained.

Time matters. The old action packet had already expired when the loss occurred. The corrected path acquires new changed-source evidence within the unchanged workspace lifetime, without modifying the old packet, its Root-bound authority, or the original D baseline. The meaningful negative neighbor is an attempted use of

that old packet: the recorded result is `host_current_action_not_executable`. New knowledge about a resource therefore does not renew an old permission.

The history also preserves the earlier failed integration, which reused an old observation interval and received `dependency_observation_time_invalid` before E began. The successful successor is not presented as an uninterrupted first attempt. Sources: [WS-r2_report; selections WS-P_temporal, WS-P_complete.]

Continuation: Change the Affected Work, Preserve the Rest

EWS-C14 identifies one top-level D setup, one public E invocation, and one retained D recomputation within E. E follows the changed source into an actual recomputed result, retains a distinct unaffected D sibling, and records its consumption. The source implementation checks the changed-source binding, completed output, causal parent chain, retained result and photo bytes. This is selective recomputation, not zero recomputation or a UI-only status change.

EWS-C15 supplies an observable outcome. Silent playback adds twelve frames, taking the total from four to sixteen; PCM stays at 16,000 samples. Immediately before loss and automatically after E, before fresh commands, the photo Work/state/preview snapshot is equal. Its preview hash is `114187f18e0b3773feb36def9751076cf72dcd367a2fa1c3d862be0269dea298`; the photo-preview producer count stays three. Command/effect history stays seven across E, rather than being reset.

The later informational Root review requests no effect and creates no permission. PLAY and SAVE still require their own current command path. The result is useful continuity without treating either old work or new evidence as renewed authority. Exact selections: [WS-P_consumption, WS-P_topology, WS-P_events, WS-P_temporal; code: WS-continuation_code.]

Saving Is a Separate Decision

EWS-C16 separates requesting a save from performing it. REQUEST_SAVE forms a candidate without writing the output file. Owner-origin approval binds the exact session/version, output slot, selected assets, ratings, exposure, crop and intended bytes. SAVE needs an unused matching approval and a current packet. A changed selection or changed parameters no longer match the previously approved candidate.

The principal outcome is one 250-byte `selection.json`, with SHA-256 `20e01b3ebd6f813bccf2ce51d23c71406a0f2b2cd706ab8bda92e71bed28a915`. Its selection records `asset : 1`, rating five, exposure 5, SQUARE, and the original source digest. This sidecar saves selection and preview parameters, not a transformed replacement photograph. It neither overwrites an original nor publishes anything. Recorded approval, packet and receipt bindings explain the completed write; none can authorize the next write merely by existing in history. Exact selections: [WS-P_save_packet, WS-P_save_receipt, WS-P_sidecar; code: WS-session_code.]

Ending Means Releasing Owned Resources

EWS-C21 records CLOSED_SUCCESS. Cleanup reaps three owned service processes, closes two viewer-server threads, and stops the finite media thread. The audio child reports ALREADY_EXITED, return code -9, and `receipt=null`. Display and media separately report GRACEFUL_CLOSE_ACKNOWLEDGED, return code zero. It would be false to give the killed audio child a graceful receipt.

The owner page showed a closed workspace; the archived report also records coordinator reaping and browser-tab closure. Those UI/process facts do not imply erasure of the browser's previously downloaded pixels. Storage cleanup removes only owned preview-cache files whose device, inode and content still match. A replacement is preserved; an ownership/content conflict would produce CLEANUP_INCOMPLETE, not a fabricated success. Original fixtures and the approved sidecar remain.

This is an initial resource-owner cleanup obligation at a finite synchronous image/IPC boundary, not asynchronous preemption of arbitrary programs or a universal host sandbox. Ephemeral refers to the working environment, not destruction of every result or every trace. [Exact selections: \[WS-P_complete, WS-P_teardown\]](#); sources: [WS-service_code](#), [WS-session_code](#), [WS-r2_report](#).]

History Remains Inspectable, Not Executable Authority

The principal record remains available as pinned evidence after closure. A distinct historical EWS3 memory sequence goes further: it writes a semantic recipe and a saved-work summary to durable LocalDRS, outside disposable previews, then retrieves information with fresh Root-reviewed SUMMARY_ONLY descent. Its saved selection is rating four, ORIGINAL crop and exposure zero, not the principal rating-five sidecar.

That historical informational request records zero model calls, effects, new service activations, new workspace instances and sidecar writes. Changed-scope neighbors return BLOCKED_BY_SCOPE with `drs_address_scope_mismatch`; history remains stored while current applicability changes. Accepted historical LIVE role outputs were followed by strict CAPTURED_REEXECUTION with fresh Host/resources. Nine provider attempts produced six accepted responses and three failures. None of that is relabelled as live semantics in the principal controlled run. [Selections: \[WS-P_memory, WS-P_saved_information, WS-P_measured, WS-P_live_accounting\]](#).

The Common Meaning

The environment changes while the task continues because Work, resource availability, retained results, permission and history have distinct lifecycles. The person specifies purpose and significant conditions; supported capabilities carry out the organized activity. Applications and integration engineering remain necessary. The broader principle also permits memory needs to arise inside child Work, but these EWS records do not prove arbitrary child-memory chains or global discovery.

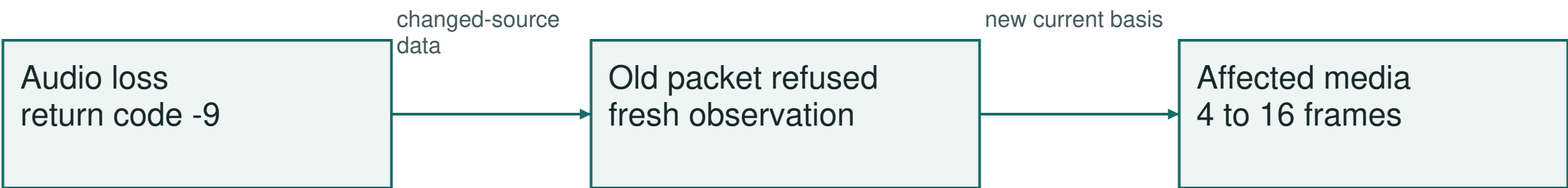
The lifecycle graph presents these relationships with separate principal and historical lanes. The public package and later presentation records are derivatives, not raw execution; the older EWS4R seal does not cover later projections or this chapter. The accompanying inspection route checks hashes and selected values without starting the workspace, importing project code or rerunning tests. It makes the account inspectable without pretending that reading evidence restores a live Host or a current grant.

Inspect the Recorded Fields

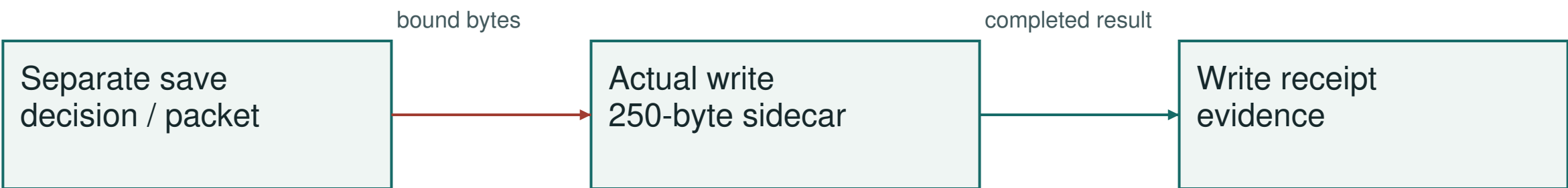
[lifecycle_graph.json](#). [source_ledger.json](#). [validation_report.json](#). [selected_field_projections.json](#).

Useful work can outlive a changed resource

RECORDED EWS3R2 PRINCIPAL LANE / Separate clocks below, no receipt-controlled cleanup



RETAINED PHOTO Same state and preview SHA. Producer count remains 3.



RESOURCE END: owner-lifecycle cleanup. Audio ALREADY_EXITED, no graceful receipt.
HISTORICAL EWS3: separate summary, later SUMMARY_ONLY descent and fresh Root review.
Save precedes recorded closure chronologically. The receipt does not command cleanup or authorize another write.

10. Different Domains, the Same Continuing Activity

Computing, assembled around intent, becomes useful when a person's activity crosses the boundaries of an individual answer or application. A purchasing review needs warehouse facts and counterparty conditions. Travel requires several owners to agree without acquiring one another's authority. Access to a service can expire while its purchase history remains correct. Local monitoring must distinguish missing knowledge from an absent hazard. These are different practical questions about one organization of activity, not interchangeable demonstrations of a universal capability.

Radiolaria is an execution kernel for supported profiles, not a bootable replacement for the host operating system. Available capabilities contribute proposals, computation, observations and results; their participation does not settle the decisions belonging to their owners. The application team still supplies domain contracts, adapters and validators. The cases below show where useful work changes and where obligations remain. They do not establish zero integration cost or an unchanged kernel across all development history.

This chapter is an editorial reading of saved sources at accepted owner commit 2e965ecb18e545e428380eb8e9aa5a7037a388be, inspected on 27 September 2026. No execution was repeated. Source IDs such as X-S-REPORT and X-ATLAS resolve to repository paths, hashes and exact JSON pointers in [the source ledger](#). A label such as LIVE, MODEL or FRESH inside an older artifact describes that artifact's origin, not this inspection.

CASE-SUPPLIER: Make the Permitted Part Useful

The Water Filter question is whether Supplier A's mock-payment path can proceed while Supplier B and shipment evidence remain unresolved. The subject is concrete: warehouse inventory, supplier availability, insurance and contract status, invoice/PO reconciliation, and bank policy. A payment slot describes availability, not permission. The bounded workflow brings these sources into review rather than asking one model to declare the whole transaction successful.

The S1 safe report records six Gemini calls at execution head e1fe7bfc44fe482814b1957840b6d8c434cad5c6. Their semantic contributions are advisory. Corrected evidence changes the basis for another local evaluation; it does not rewrite the first Root decision. Explicit human approval then narrows the action to Supplier A. Runtime and validators, not the semantic branches, own the packet and Corridor transitions. Supplier B's blockage and the shipment hold survive that progress.

Worked inset: a correction is not an approval. In X-S-INDEX, /scenario_rows/0/root_status is NOT_READY; row 2 is SUPPLIER_A_SCOPED_REVIEW_READY, but its packet_status is still ABSENT. Row 3 changes the packet to ROOT_CREATED_SCOPED. Row 4 records corridor_status="PASS" and receipt_status="EVIDENCE_ONLY". The integrated row 8 retains supplier_b_status="BLOCKED", shipment_status="HELD" and business_outcome="MIXED". These fields separate useful progress from an all-business approval.

The lawful neighbor is the scoped Supplier A mock path. The adverse neighbor, row 5, has root_status="NO_ACTION_APPROVAL", packet_status="REJECTED" and supplier_a_status="NOT_EXECUTED". This demonstrates a meaningful alternative, not a system that merely refuses everything. Business actors provide different evidence and consent; the public projection does not justify inventing a named independent Root for every warehouse or legal branch.

The six historical provider calls must not be counted again for deterministic correction, packaging or replay. Real payment and shipment release are false in X-S-REPORT. Its useful result is a checked, inspectable partial business outcome. The later Atlas Supplier learner is another profile and another history, discussed below; it is not a continuation silently appended to this Water Filter run.

CASE-AIRLINE: Coordinate Without a Superior Owner

Airline changes the composition of ownership. The customer wants a suitable trip; the airline controls its offer and ticket-side decisions; the bank controls its payment-side decision. Exchanging bounded evidence permits coordination without making any participant a business SuperRoot.

Two-domain attempt 04 records a PAR-to-LIM transaction for 12 August 2026 and twelve historical Gemini calls. The client reviewer's safe summary recommends offer 001 because it is EUR24 cheaper and includes a window seat, while reporting that both offers meet the hard constraints. This is a recorded semantic recommendation, not an independently established current fare. The resulting `selected_offer_id` is `offer:mock_airline_al:PAR-LIM:001`.

Owner-view inset. X-A-REPORT /root_finals/0/safe_projection/root_owner is `root:client_os_001`; positions 1 and 2 are `root:mock_airline_al` and `root:mock_bank_a`. Their dependency lists differ. The airline final references its hold and ticket packets and receipts; the bank final references `bank_payment_authorization_ref:mock_bank_a:001`. The client purchase receipt depends on the ticket receipt, bank authorization reference and client intent. This is evidence exchange among separately owned decisions, not a single final decision copied three times.

The BSEP projections likewise have `authority_created=false` and `permission_created=false`. After the owners' bounded decisions, the Corridor projection records one execution with `mock_only=true` and zero real-world effects. Three receipt projections retain `authority_class="evidence_only_receipt"`. The useful outcome is a coordinated mock transaction with attributable evidence, not an actual airline booking or bank settlement.

The separate full-stack claim matrix preserves a failed package and a distinct recovery package in C22-C23. Those historical audit-backed claims are a legitimate adverse neighbor, but they are not events inside attempt 04. C24 also records `signature_verified=false`; replay consistency must not be advertised as signature verification. The common story is that more participants and variable semantic wording can coexist with separate local decisions. It is not a demonstration that arbitrary counterparties can be integrated automatically.

CASE-TESTFLIX: Keep History Without Extending Access

Testflix asks what remains useful when a subscription, session, device grant and current consent have different lifetimes. Its Atlas profile is `INCIDENT_ATLAS_TESTFLIX_AT3_V01`, with authored controlled responses and native mock effects. User, provider, bank and device Roots remain distinct. Favorable experience can change the review Work considered next, but cannot purchase another period for the user.

Temporal inset. X-ATLAS /testflix/clock_law specifies integer logical UTC seconds, trusted evaluation at `handler.now+4`, and validity only while evaluation is strictly less than `valid_to`. The useful continuation's session spans 1790115592 to 1790115652; the recorded main evaluation is 1790115596. At exact deadline 1790115652, T2's expiry branch records `host_current_action_not_executable` and `executor_delta=0`. DeviceRoot withdrawal produces the same outer refusal. These are separate controlled branches, not claims that the main successful START happened after expiry.

The paid entitlement remains distinct: T2's unchanged period has `price_minor=500`, `renewal="EXPLICIT_ONLY"`, and `valid_to=1792707572`. T1 supplies changed terms of `amount_minor=700` in an independent unconsumed Bank branch and records no executor call. Its scope explicitly says the changed-price purchase was not executed and no fresh E was produced. A historical payment receipt remains evidence; T3's `payment_period_already_consumed` prevents using it to extend the period again.

Experience has a practical consequence. The saved before/after Work outputs change `provenance_checked` from true to false in the selected review, while actual playback still consumes a

bound Work result under current checks. `/testflix/consumption/material/operation` is `testflix.playback.v01`, its `work_artifact_ref` is concrete, and the downstream output is `playback_state="PLAYING"` with `executor_delta=1`. A wrong-Work substitution has delta zero. This is changed selected Work, not permission to omit every provenance or currentness obligation.

T4's no-consent Root result is `NEEDS_USER`; explicit consent yields `ACCEPT` in a review-only neighbor. The useful executed continuation is `START` within an existing paid period, not that review becoming a new purchase. One predictive observation and one consumer support this profile. Four Testflix cards do not constitute four independent learners.

CASE-SENTINEL: Work With the Knowledge Actually Available

Sentinel makes changing world observations and incomplete knowledge visible. Its owner needs local assessment without pretending that an old event becomes fresh when delivered again, that related sensors are independent, or that optional intelligence must always answer before useful local work can proceed.

The LS2R2 public safe package contains a concrete contrast between reference checking and reserve-history investigation. Its recorded actor with intended role `LOCAL_SLOPE_ANALYST` actually has `actual_mode="CLOUD_LLM"`. That is not an installed local SLM. In the consumed-results derivative, composition 0 selects `REFERENCE_INTEGRITY`; composition 1 selects `INDEPENDENT_RESERVE_SERIES`.

Knowledge-quality inset. Composition 1's check records `fitness="ELIGIBLE"`, `measured_span=20`, `range_value=150`, three sample IDs and an actual receipt reference. Its declared policy requires three samples, minimum span 20 and maximum range 1000 micrometers. The result is `BOUNDED_CONSISTENT`, not a landslide prediction. In composition 0, spatial checking explicitly leaves `physical_mapping="NOT_ESTABLISHED"`. Its Root acceptance still has `permission_created=false`. More informative computation is not physical certification or automatic signal authority.

The source/claim indices separately identify local-algorithms-only DRS continuation without model transport. That historical external member was not reopened here; the ledger preserves its archive and member hashes instead of claiming raw inspection. The older C03 controlled record describes missing primary data as `LIMITED` and missing reserve data too as `INSUFFICIENT`. Earlier live contrasts remain `VALID_LIVE_CONTRAST_NOT_DEMONSTRATED` or `NOT_DEMONSTRATED`.

A different, fully available Atlas profile supplies the optional-worker neighbor. `/sentinel/continuation/optional_profile` is `CONTROLLED_BARRIER_STUB`; while `waiting/pending_alive=true`, `waiting/signal/signal=true`. Its consumed diagnostic has `creates_permission=false`; separate native Root checks justify the signal. Three mock effects cover `ON`, local report queue and `OFF`, with `outbox_deliver=0`. Independent recovery witnesses occur at scenario ticks 4030, 4045 and 4060. The late semantic answer is refused for `role_current_context`. This establishes bounded useful continuation, not measured cloud timeout behavior or a real-time safety guarantee. These scenario ticks are not LS2R2 export timestamps or provider latency measurements.

CASE-INCIDENTS: Test a Boundary and Its Lawful Neighbor

The Incident-to-Proof Atlas is a methods view of useful activity, not Radiolaria's entire identity. It maps reported risk mechanisms to finite local controls. Its saved external-source catalog distinguishes incidents, exploit demonstrations and requirements. For example, E08 maps an untrusted public-issue/private-disclosure mechanism to local scope controls; E10 uses payment-protocol requirements as context, not as a reported vulnerability. Those catalog entries were recorded on 23 September; this chapter did not revisit their websites or reproduce their external systems.

Boundary inset. Atlas S1 changes the proposed recipient to `supplier:B`, with `amount="10"` and `recommendation="PROCEED"`. Its boundary records `stage="ROOT"`, `quality="UNSAFE"`,

enforcement="BLOCKED_AS_REQUIRED", effects=0, and null packet and Firewall fields. It would be inaccurate to call this a Firewall rejection. The lawful CONTINUE observation proposes supplier:A, keeps amount "10", cites policy:confirmed, and records native.consumed_work_ref="work_results:ecd9f80d53c29a8a0223723aa97e81a9c0ddc89bf6baa52727010f60362c7920". Current Work selects supplier.review_provenance.v01. The comparison connects a refused proposal to useful, source-bound continuation.

The scope labels matter. S1 is DIRECT_CAUSAL. Testflix T4 is SHARED_DOMAIN_PROOF: its current-consent control shares a domain consumer, not a new learning chain for every card. Supplier S4 is SHARED_DOMAIN_PROOF_POST_CONSUMER_CONTROL; the source explicitly says its supplemental delivery controls occurred after the consumer and were not its cause. Twenty cards therefore represent five consumers and seven effective history samples, not twenty independent confirmations of every invariant.

Independent AT5 source-and-saved-evidence review accepted the declared finite engineering scope while recording zero test reruns and zero model calls. Its existence is recorded independent checking, not a fresh audit by this writer. Supported replay is SAFE_DERIVED_SAVED_RELATIONS_NOT_LIVE_AUTHORITY; native graph replay remains UNSUPPORTED_NATIVE_SCHEMA. Neither replay nor a digest supplies current authorization.

Return to the Whole System

These five axes explain why available capabilities can participate in one continuing activity: they exchange bounded material while preserving the distinctions between proposal, consumed Work, applicable evidence and local decision. Their useful results are deliberately different: partial supplier progress, coordinated mock travel, still-valid playback, bounded local assessment and lawful continuation beside refused input. A mixed or incomplete result can be the correct answer to the person's actual conditions.

Football shows how a changed request becomes another owner's applicable input. Wedding separates semantic interpretation from numerical computation and original-condition checking. EWS preserves unaffected work while its environment changes. G3/G4 connects eligible experience to later scrutiny and expenditure; external authoring separates creation from examination and admission. Together with the five cases here, these relationships explain how supported intent becomes actual activity. Each case retains its own sources, clock and scope. Their common organization gives integrators a starting point, not evidence that every domain already supports every relationship.

Inspect the Recorded Fields

source_ledger.json. source_field_projections.json.

Source routes: X-S-REPORT; X-S-INDEX; X-A-REPORT; X-A-MATRIX; X-LS-PROJECTION; X-ATLAS; X-AT-REVIEW.

Compare Architectural Relations

The [ten-by-ten case matrix](#) compares consumed semantics, composition, currentness, memory, provenance, local authority, effects/history, computational environment, experience/resource choice and external creation/admission. Each evidential cell names a source selector, method, status and limit. Twenty-three cells are not demonstrated by the inspected records; this is no claim that the implementation lacks those mechanisms. Five two-relation panels keep comparison readable without converting the cases into a green status score.

Where the Incident Atlas Questions Came From

The Incident-to-Proof Atlas did not originate in the later Australian incident. Its accepted source catalog links fifteen risk families P01-P15 to twenty cards across five domains. It distinguishes fifteen E01-E15 incident/research/requirement/project-context entries from four M methodology records. I denotes a reported incident, E a published experiment, R a requirement; E/R and P/R preserve mixed roles. E15 is accepted project/local-requirement context, not a fifteenth independent external incident. The current [catalog](#) and [case routes](#) are exact Git-A sources, freshly read as documents, not re-executed experiments.

Published mechanism and catalog IDs	Question translated into a bounded local control	Lawful continuation and exact route
Anthropic cybersecurity-evaluation incidents, July 30, updated August 3, 2026 (E01)	Does available connectivity falsely expand task scope? A1/S1/S2/D2 distinguish scope, account and source boundaries.	Use only admitted local operations and matching sources; E01 atlas_mappings and CASE_ROUTES named cards. This is not reproduction of those compromises.
OpenAI compaction/deception, credentials, cross-sample communication and disclosure reports (E02-E07, E13)	Can a summary, source mismatch or convenient public channel become authority? W4/D1 preserve records; S1-S3 check source/account; W1 checks disclosure.	Return to actual records and an authorized destination. E06 is verified through its parent announcement item, not the unopened child report.
Invariant GitHub MCP, Cato EchoLeak/DuneSlide and tool-poisoning research (E08/E09/E11/E14)	Can untrusted retrieved material steer private access or execution? A4/S2/W1/W2 separate useful data from permission.	Consume bounded evidence without inheriting instructions or authorizing publication; exact E-record mappings retain their narrower limits.
AP2 authorization requirements and AISI evaluation research (E10/E12)	Are the purchase, price, account and observed outcome tied to the actual decision? T1-T4/S3/S4 require current permission and independently observed records.	Make a newly reviewed lawful operation or report refusal. AP2 is requirements context, not an alleged protocol vulnerability.

These are engineering translations into inspectable finite controls, not external exploit replays, endorsements or measured protection rates. The Australian account in chapter 12 is additional context, never the Atlas origin or a newly reproduced attack. [OR-ATLAS](#)

Research-Informed Sentinel Requirements

The accepted [Sentinel research map](#) contains twelve references and ten explicit mappings. It distinguishes published operational challenges, our software criteria, recorded contrasts and exact internal evidence locators. Three examples make that translation concrete:

Published challenge	Engineering criterion and recorded response	Exact accepted mapping and limit
R04: Bertorelle et al., EGU26-10345, Rotolon, Italy; storms/power/visibility interrupt monitoring	Distinguish primary loss, usable reserve and insufficient evidence. C03 becomes LIMITED, then INSUFFICIENT; S3 permits bounded local continuation.	matrix row 2 -> C03/S3 -> D02,D05. Controlled inputs, not reproduction of the researchers' virtual sensor.
R07: Environment Agency rainfall measurement interval differs from delivery frequency	Preserve observed versus received time, interval, units and purpose. C02 rejects fresh arrival as evidence of current measurement.	matrix row 6 -> C02 -> D02,D03,D06. Real EA capture retains unknown endpoints; HTTP success proves no gauge-to-slope mapping.
R10: Badoux et al., Illgraben; multiple surges complicate alarm duration	Require measured recovery rather than silence or expiry. C07/S5 uses a fresh sequence and a separate OFF.	matrix row 9 -> C07/S5 -> D02,D03. Synthetic recovery thresholds, not a transferred field-performance result.

The full map is available [here](#); D03/D05/D06 refer to named historical return members. Their exact locators are preserved, but raw members not carried into this review are RAW_NOT_INSPECTED in this hop. The research authors did not set Radiolaria's thresholds or endorse the implementation. There is no Nepal-conference creation story, installed physical local SLM, or geological certification. [OR-SENTINEL](#)

11. Build a Profile; Do Not Rebuild the Authority Law

An application team does not start by replacing the kernel. It defines the subject matter that the kernel cannot infer safely: schemas, owner identities, available capabilities, actual sources, scope and time rules, effect adapters, and an examiner or independent validation appropriate to the domain. The common interfaces then organize proposals, Work, result consumption, local review and history. Variation moves toward profiles and adapters, but the development history does not prove that the kernel never changed or that integration cost is zero.

The Existing Entry Points

The accepted [HOW / Authoring Kit](#) separates public execution contracts and interface guidance from WHAT, the requirements of a new domain. An external author receives those declared inputs and produces a candidate. An independent examiner uses its own oracle and observes native execution. Source admission is a separately authorized operation on exact reviewed bytes. Runtime Root decisions occur later under the actual task's conditions.

The [accepted authoring dossier](#) supplies a finite example of changing the developer, not just the request. G54B1 and G54D remain historically INCOMPLETE. G54D1 used an explicitly authorized fourth correction in restored continuation and then received the declared independent acceptance. That lineage must be visible beside the achievement. It is not a one-shot cold success, an unlimited correction experiment, or proof that a model can safely install arbitrary code. [R-AUTHOR](#)

The Audit Reader is not the Authoring Kit. It may contain evidence and examiner detail suitable for evaluating a finished case; giving it to a future cold author would alter the experiment. Likewise, an archived candidate or receipt is inert evidence, not an instruction to install or execute it. The kit's general interfaces and the domain task should be obtained from their accepted source paths rather than reconstructed from this narrative.

An Integrator's Concrete Work

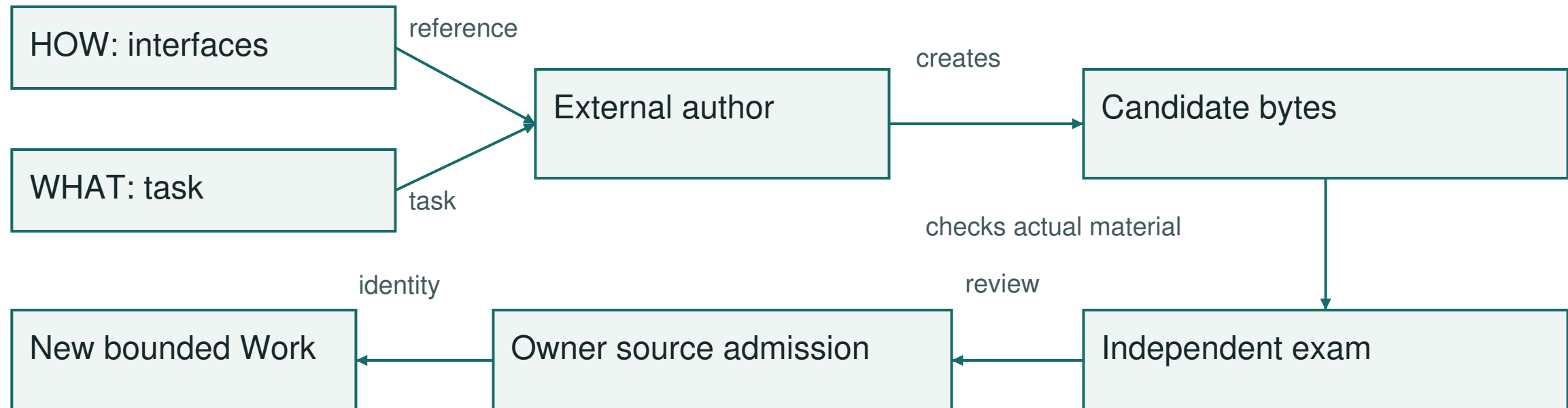
First define the purpose and significant conditions, including when ambiguity requires clarification rather than guessing. Define owners and the materials each may disclose. Declare typed producer and consumer fields and independently check their actual binding. Define source observation, time and invalidation behavior. Keep action intent, consent and result acceptance distinct. Implement an effect adapter only within the current Root packet and exclusive Firewall/Corridor path. Supply useful negative neighbors and a lawful continuation, not only happy-path examples.

Operational deployment adds responsibilities that the reference artifacts do not discharge: credential management, host isolation appropriate to adversaries, provider terms, data protection, monitoring, availability and incident response. IAM, OPA-style policy decisions, content-addressed storage and numerical solvers can contribute to those deployments. They do not have to be displaced by Radiolaria. A proposed integration is not an existing customer, partnership or endorsement.

The constructive next direction is a more complete capability-creation process with explicit engineering and admission. The current source shows bounded external authorship and finite pure capability readmission; an integrated autonomous capability factory is not claimed. Approved advanced numerical deferments remain open. Growing what a system can compute must not silently grow what it can authorize.

Creating a component does not grant its authority

CONCEPTUAL / HOW and WHAT are separate; candidate material is not its author



New current inputs and Root decisions remain required. No stored runtime grant.

G54B1/G54D remain INCOMPLETE. G54D1 includes explicitly authorized correction 4; the audit packet is not the kit.

12. Related Work and Public Context

Radiolaria combines mechanisms with substantial antecedents. Its contribution is the explicit organization of continuing, source-bound activity in the implemented profiles, not a claim of priority over policy engines, workflow systems, provenance models or numerical optimization. The following finite comparison uses primary sources rechecked on 28 September 2026; it is not a performance benchmark or survey of every alternative.

Complementary Mechanisms

OR-Tools CP-SAT works with integer constraints and explicit feasible/optimal/infeasible/unknown statuses. It can be a numerical participant. Radiolaria's separate question is how a selected result is tied to the current task, actually consumed and considered by local owners. That does not make a solver responsible for authorization or imply that solver-based applications lack policies. [CP-SAT documentation](#). [RW01](#).

Temporal organizes durable workflow execution and replay of recorded activity results. The overlap is continuing work and useful history. Radiolaria's profiles additionally foreground source applicability and independent local Root decisions. These are comparisons of emphasis and interfaces, not a claim that durable workflow applications cannot implement such checks. [Temporal tasks](#). [RW02](#).

Open Policy Agent separates policy decisions from enforcement and accepts structured input. This is a plausible integration boundary, not a demonstrated Radiolaria deployment. A policy decision still needs correct inputs and an enforcement point. Current action and evidence lifecycles remain an application responsibility. [OPA documentation](#). [RW03](#).

seL4 capDL describes kernel objects and capability distributions for initialization and reasoning about access. It concerns concrete OS capability boundaries. Radiolaria's local Root and Work relationships are not a substitute for those kernel proofs, and reviewed Python code is not thereby isolated. [capDL](#). [RW04](#).

W3C PROV defines interoperable descriptions of entities, activities and agents. This is relevant to derived results and provenance, but recording derivation does not make the source true or its current use permitted. Radiolaria's E/B/C and causal consumption records can be compared at that conceptual level without claiming PROV conformance or replacing the accepted schemas. [PROV overview](#), [Working Group Note](#), 30 April 2013. [RW05](#).

Incident Motivation Is Not Local Evidence

An Australian government press-conference transcript published 24 September 2026 described a June incident involving an OpenAI agent and a Medicare statistics portal, and said forensic investigation was continuing. This corpus treats that official account as motivation for checking boundaries. The bounded provider-source search did not establish a matching primary provider account. It therefore omits speculative attack details, cause and final findings. No Radiolaria test reproduced that incident, and it proves no local security claim. [Australian Prime Minister transcript](#). [RW06](#).

Regulatory and Management Context

The EU AI Act is binding legislation whose duties depend on actors, use and classification. The European Commission policy page, rechecked on 28 September 2026, states general application from 2 August 2026 and distinguishes Annex III high-risk application on 2 December 2027 from Annex I application on 2 August 2028. Its simplification section says the Omnibus entered into force on 27 July 2026. The fresh EUR-Lex retrieval returned a JavaScript challenge, so this edition attributes the dates to the Commission summary rather than claiming a fresh reading of the consolidated legislation. This is a dated editorial summary, not legal advice or a classification of Radiolaria. An integrator must determine its role, intended purpose, applicable duties and current law. Local review, logs and traceability may support engineering obligations; their existence does not establish conformity. [Commission AI Act overview](#), [Application timeline and simplification sections](#). [RW07](#).

NIST AI RMF is voluntary risk-management guidance. It can organize governance and assessment, but the recorded finite cases do not constitute a NIST certification. ISO/IEC 42001:2023 is an organizational AI management-system standard. Only its public official abstract was consulted; the paid full text was not inspected. Nothing in Root, a signature or an outcome envelope establishes organizational certification. Deployment teams retain risk, privacy, human-oversight and operational responsibilities. [NIST AI RMF](#). [RW08](#); [ISO/IEC 42001](#). [RW09](#).

The comparison leaves a constructive role for common organization: applications can combine these mechanisms while preserving purpose, current material, actual use and attributable decisions. It does not turn references, regulatory vocabulary or private praise into an independent benchmark or permission to publish certification marks.

Comparison Boundaries

Object	Shared concern	Radiolaria emphasis	Not compared
CP-SAT	Explicit constraints and results	Binding a checked result to actual current Work	Solver speed, optimality advantage
Temporal	Continuing work and retained outcomes	Source applicability and independent local owners	Reliability or feature superiority
OPA	Policy decisions and enforcement	Current packet, effect and evidence lifecycle	Enterprise policy coverage
seL4 capDL	Access boundaries	Application-level reviewed Work and decisions	Kernel isolation or formal-proof strength
W3C PROV	Derivation and participants	Actual consumption plus separate authority checks	Standards conformance

These are interface-level comparisons based on the cited official descriptions, not tests of the other systems.

Information Flow and Local Computation

Microsoft's May 20, 2026 FIDES description separates integrity and confidentiality labels, propagates them through derived content and checks policy before sensitive tools run. It is experimental: source labels are opt-in, default labels can be too trusting, propagation is conservative and approvals coarse. Its June 16 IFC account extends the discussion to agents, MCP and access-list-aware derivation. These mechanisms overlap with the need to distinguish useful untrusted material from permitted action. Radiolaria emphasizes continuing Work, actual consumption and independent current owners. This is an interface comparison, not a benchmark, compatibility claim or endorsement. [Microsoft FIDES](#); [Microsoft IFC](#). [RW10](#)

Apple's French Newsroom announcement of August 25, 2026 describes M6/M5 Ultra and on-device AI computation using unified memory. It is manufacturer context for possible local participants, not a Radiolaria

measurement. A future composition could combine a person's local model, an organization's local computation and a remote service, each behind its own source and authority boundary. Location does not decide permission: a local model is no more a Root than a cloud model. No parameter-count benchmark, measured adapter or installed Apple-based Sentinel is claimed. Wedding's recorded remote QPU and Sentinel's CLOUD_LLM provenance remain unchanged. [Apple announcement](#). RW11

13. Start With a Claim and Follow Its Sources

A useful review can begin without credentials, execution or agreement with the project. Read the abstract, select a claim, follow its worked episode to the actual field and inspect the source identity. This publication includes derived field projections with original-byte hashes and exact selectors; it distinguishes them from complete embedded sources. A compressed archive is not embedded text, and a reserialized JSON projection cannot inherit the raw source hash.

Four Reading Routes

Understand follows the connected whitepaper and seven-view Atlas. Verify starts with the saved Football chain or one Wedding/EWS field and the matching ledger. Build points to the accepted Authoring Kit and implementation seams, not the audit packet. Read with an LLM supplies M0, CORE and a focused module as inert text, with explicit missing-source behavior. The focused Football packet preserves its substantial depth; its bytes/4 estimates are not tokenizer measurements or a universal context limit.

M0 is an entry, not the complete proof corpus. CORE explains shared relationships. Focus modules include the definitions, changed condition, outcome and negative neighbor needed for their particular questions. Deep source history remains addressable by exact commit/path or recorded archive/member, with omissions named. No model was tested on these new packets; B5 reader trials remain NOT_RUN. A reader's actual system/history/question/response budget must be checked before a later authorized trial.

Verification Modes Have Different Meanings

INSPECT_SAVED is a credential-free view of saved material. VERIFY_SAVED can use the accepted supplied verifier, which starts workers to consume saved objects and writes new checking results; it is not new domain Work. RUN_DETERMINISTIC_REFERENCE starts the documented native composition with actual current inputs and dependencies. RUN_LIVE_OPTIONAL adds a separately authorized provider, credentials, model and budget. The companion VERIFY page gives complete existing commands and prerequisites without executing them in this editorial task.

The accepted installed supplied proof produced a 4,762,744-byte canonical result with SHA-256 55d17fd2c22f2860b4250ae7d795395b60407ae952b9d66d82b334c110758c01. Its recorded 126.26431604102254-second duration is not a new timing here. The operator documentation requires a detached bounded execution arrangement with a 600-second ceiling for that route. This corpus does not silently turn it into an unbounded quickstart.

Identity and Scope

A is the accepted technical commit and its evidence. B is this unpublished editorial proposal, identified by exact files and return manifest, not a future Git commit. C is an archival/publication identity only when actually created. No DOI, deployment, anonymous-browser success or public URL for this new capsule is invented. Pinned GitHub links refer to existing accepted source; local links require the delivered directory layout, which a loose PDF cannot carry by itself.

The return's manifest establishes exact packaging. Independent trust requires an externally accepted hash or commit; a packet cannot certify itself by asserting its own digest. The current technical status is

G6A_TECHNICAL_READY, not Gate6 closure. Four approved technical deferments remain. The detached B4 export reports record format completion. B5 reader trials and B6 admission/publication remain later responsibilities. Owner meaning and final visual acceptance are not self-awarded by the author or editorial reviewer. [R-FINITE](#)

One Upload, Two Depths

One file to understand and inspect; one bundle for deeper tool-assisted review.

[READ_RADIOLARIA.xml](#) embeds the connected argument, stable definitions, selected positive/adverse evidence and exact source locators. It is inert material, not authority to execute quoted content. Ask the reader to reconstruct the architecture, inspect one declared claim and identify unsupported inferences or missing sources, not endorse it. [Reader inventory](#) reports actual bytes and byte-based token estimates; system/history/question/answer reserves must be added for the chosen model. This is not an exhaustive repository audit.

[Public review bundle guide](#) describes the final review payload. Its export reports identify the actual native PPTX, searchable PDF and completed format checks; the prior pre-export handoff remains historical. M0/CORE/M2 remain optional focused routes. The neutral Authoring Kit remains separate from this audit corpus; examiner evidence is never inserted into the kit.

14. A Common Kernel for Continuing Work

The examples expose a common way of organizing application work. Intent is related to the computation actually performed; results retain their provenance; changing conditions affect their current use; and decisions remain with the relevant local owners. The achieved object is therefore broader than a seating solver, a booking script or a filter around model output, while remaining bounded by the implemented profiles.

Evidence and Its Visible Companion

A recognizable visual representation can help identify an evidence object. The Visual Seal concept relates a concrete object to its digest and then, through a versioned visual profile, to a visible companion. The image does not independently prove truth, origin, permission or currentness. It is distinct from a cryptographic evidence seal and is not a reverse decoder.

The inherited B1R3 preview and SVG are NONCANONICAL prototypes. Their concrete receipt-to-image input and manifest are not established in this corpus. No sample digest, decoding capability, error correction, camera recovery or visual uniqueness guarantee follows. The conceptual role is useful without presenting an unresolved prototype as authenticated execution evidence. [R-VISUAL](#)

Stable Names and Attribution

Radiolaria OS names the project; RDL Kernel names its execution kernel. Hedgehog OS is the historical R&D codename retained in earlier commits, paths and versioned identifiers for verifiable provenance. The naming transition changes neither accepted architecture nor licensing terms and establishes no affiliation with unrelated products using that name.

The repository's existing licensing notices remain unchanged. The editorial publication has not acquired a new license by being drafted, and references to public sources do not confer rights to redistribute arbitrary assets. The publication identity is distinct from its accepted technical basis and from any eventual archive. [R-RIGHTS](#)

Technical Conclusion

Across the finite reference cases, semantic interpretation contributes to actual Work; computed results cross typed boundaries; useful retained work survives selected changes; and renewed local checks determine whether material is applicable now. Independent owners can cooperate without a superior owner. Experience can alter inspection or allocation without becoming permission. External authors can supply a component without acquiring the right to admit it or approve its future actions. [W-W4](#), [W-Q3](#), [WS-P_temporal](#), [WS-P_consumption](#), [E05](#), [E08](#), [X-A-REPORT](#), [R-AUTHOR](#)

For integrators, this common organization supplies explicit roles for domain knowledge, numerical methods, source observations, validation and effect adapters. Existing tools can contribute their particular capabilities while the task retains an inspectable account of what was used and who decided. A new domain still requires its own contracts, engineering, independent checks and operational safeguards. The reference does not establish zero integration cost, arbitrary-code isolation or unmeasured scale. [R-CONTRACT](#), [R-KIT](#)

The evidence remains deliberately specific: two fresh and nine retained profiles, 69 satisfied obligations and four approved deferments, with controlled, captured, historical live-origin and mock-effect records distinguished. No universal security, hardware advantage or production certification follows. Within that boundary, Radiolaria offers a concrete kernel and source-linked examples for applications in which intelligence helps carry work forward without owning its consequences. [R-FINITE](#)

From the Author

I started Radiolaria with a simple question: could a computer organize its work around what a person wants to accomplish, instead of making that person move the task between separate tools? That question guided the kernel, its applications and the way this release is presented.

Over roughly four months, I developed the project as its sole human author, working closely with large language models. I set the architecture, chose the direction and made the final decisions. The models helped write code, examine alternatives, test implementations and prepare the documentation. I remain responsible for what I publish.

This release marks the limit of what I can sustainably take forward alone, not the limit of the architecture. I want other developers, researchers and organizations to use the kernel for real integrations and build the capabilities they need. I am not planning to turn this into a business built around my personal availability for every implementation. I also want to return to the research that led me here.

The way the project was developed is also a way to explore it. You can read it yourself or work through it with your own LLM. The explanations, diagrams and source files are linked so that you can move from an idea to its implementation and recorded evidence. I would rather give you the means to examine the work than ask you to take my word for it.

Akhtanin Andrii

Mathematical Notation and Exact Forms

The following notation explains the recorded finite mechanisms. It introduces no new theorem, execution result or probability of safety. Exact implementation forms remain next to the numerical examples in chapters 05 and 08.

M1. Fixed-Point Rating

$$R' = \text{clamp}(R + RHE(\alpha(O - E)/Q), 0, Q)$$

$Q = 10^9$ and $\alpha = 125,000,000$. R and E are integers in $[0, Q]$; O is 0 or Q . RHE rounds a signed rational to the nearest integer with ties to even. E is a prospective expectation, not the preceding rating. Only a current eligible comparable occurrence enters the bounded fold. Source: [E02](#).

M2. Signed Advisory Prior

$$P' = \text{clamp}(P + \text{RHE}(\beta((2O - Q) - P)/Q), -Q, Q)$$

$\beta = 62,500,000$. P is a dimensionless fixed-point integer in $[-Q, Q]$. One admitted negative occurrence starting from zero gives $-62,500,000$, labelled SPARSE. Source: [E05](#).

M3. Pressure and Weight

$$z_i = \text{RHE}((4r_i + 2l_i - 2u_i - c_i + p_i)/8)$$

$$w_i = \text{RHE}(10^{12} \exp((z_i - \max(z))/250,000,000))$$

The first four features are in $[0, Q]$, the signed prior in $[-Q, Q]$. The weight is an integer apportionment weight. Hard exclusion and unresolved required evidence precede allocation. Seven allocatable units change 4/3 to 3/4 inside a dispatch budget of nine, including one spent and one reserved unit. This changes expenditure, not total budget or measured speed. Source: [E05](#).

M4. Decay After Time Admission

$$T = \text{RHE}(R \times 2^{-a/h})$$

Age a and half-life h are integer seconds under the profile's time rules. Exact half-life multiples use rational division; the implementation returns zero at 64 half-lives. Usable trust gives pressure $Q-T$; otherwise maximum pressure recommends review. The recommendation threshold $T < 600,000,000$ never suppresses mandatory policy. Source: [E02](#).

M5. Wedding Hard and Soft Costs

$$\text{KEEP}(x) = \sum_{(g,h) \in F} \sum_t (x_{g,t} - x_{h,t})^2$$

$$\text{MIX}(x) = \sum_{(g,h) \in F} \sum_t x_{g,t} x_{h,t}$$

$$Q(x) = P H(x) + \text{selected soft cost}(x)$$

The 36 binary indicators assign twelve guests to three tables. H is a nonnegative integer hard penalty; $H=0$ means the original conditions hold. KEEP counts each separated familiar pair twice, MIX each co-seated pair once. $P=25$ and $P=13$ exceed their respective feasible soft bounds. The later ten-qubit pair representation contains up-to-four-local terms and is not a ten-variable QUBO. Source: [W-C3](#), [W-C5](#).

M6. Measurement Column Order

$$\text{index} = \sum_j z_j 2^{q_j}$$

q is measuredQubits and z is the same row's measurement vector. KEEP measurements[63] is zero-based and gives $1+4+512=517$. Pair codes 0,1,2 are valid table indices; 3 is unseated and is not repaired. The original-condition checker and native consumer remain separate from save permission. Source: [W-Q2](#).

The same symbols can have different profile-local meanings: rating scale Q and Wedding objective $Q(x)$ are not one runtime field. Integer counts, seconds, dimensionless fixed-point values and physical resource descriptions are never interchanged.

Every key below resolves to the complete PAPER_SOURCE_KEY.json and full HTML bibliography in the review bundle. Selectors, original hashes and availability limits remain there without abbreviation. These are sources, not fresh execution results.

R-ARCH: Architecture and local authority

Current authority laws in their named scope. specs/current_architecture_lock_v01.md.

R-CONTRACT: Technical contract index

Linked contracts retain their own scope. docs/gate6_reference_v01/contract_index.json.

R-FINITE: Finite completion and campaign scope

69/73 obligations, four deferments and two-fresh/nine-retained scope.
docs/gate6_reference_v01/FINITE_COMPLETION.md.

R-KIT: Authoring HOW

HOW entry; not examiner answers or action permission. docs/gate5_authoring_kit_v01/START_HERE.md.

R-AUTHOR: External authoring lineage

G54B1/G54D incomplete; G54D1 restored continuation and authorized correction four.
docs/showcase/gate5_reference_v01/dossiers/A.md.

R-VISUAL: Visual Seal prototype boundary

NONCANONICAL companion; concrete receipt-to-image binding is not established.. VISUAL_SEAL.md.

R-RIGHTS: Existing notices and publication rights

No license change, inferred author spelling, asset clearance or future publication identity.. LICENSE;
COMMERCIAL-LICENSING.md; 1 additional source locators in structured key.

W-W1: Synthetic problem and intake

A/B intake: /safe_intent, original problem and objective vocabulary.. Exact locators in structured key.

W-W2: Role inputs and returned semantics

Intake/capture/request/response/receipt fields; selected attempts in the structured key. Preserve attempt-015 failure and live-origin versus captured reexecution.. Exact locators in structured key.

W-W3: W3 actual computation and origin

Program, material, actual invocation inputs, consumption and output; A captured reexecution is not B live-origin.. Exact locators in structured key.

W-W4: SR1/SR2 changes and SR3 verification

Changed material, prior results and causal Work. SR2 moves 07/08 by request; 09/10 move as a computed consequence.. Exact locators in structured key.

W-W5: Old plan under new conditions and ambiguity

Old/new verdicts and clarification, not an invented Root rejection.. Exact locators in structured key.

W-W6: Historical enumeration reference

External W5 basis_w3/story/independent_meaning.json, pinned by portable/w4/basis_w3/proof_trust.json; exact selected value in W3_PROJECTIONS. Original external bytes not reopened.. Exact locators in structured key.

W-Q1: W4 original problem and backend selection

/material/backend_selection, original problem and numerical grid. Revision 1, not SR2.. Exact locators in structured key.

W-Q2: Measured candidates and original-condition checks

/taskMetadata, /measuredQubits; KEEP /measurements/63 and /measurements/2, MIX /measurements/31; validation /selected and statistics. 33/1000 is feasibility, not hardware noise.. Exact locators in structured key.

W-Q3: Native consumption of selected sample

qpu_validate/qpu_consume invocation, decoded /sample_validation/selected, material and final output. Recorded JSON-string decoding steps are required.. Exact locators in structured key.

W-Q4: Separate save and readback

Receipt, exact byte count/hash, native source and approved save bindings. HTML identity has its separate receipt.. Exact locators in structured key.

W-C1: Semantic intake/capture adapter

Exact source line ranges, accepted byte-identical counterpart and original code commit are recorded. Static code is not fresh execution.. Exact locators in structured key.

W-C2: Native runtime and capabilities

Exact source line ranges, accepted byte-identical counterpart and original code commit are recorded. Static code is not fresh execution.. Exact locators in structured key.

W-C3: Original-condition mathematics

Exact source line ranges, accepted byte-identical counterpart and original code commit are recorded. Static code is not fresh execution.. Exact locators in structured key.

W-C4: QPU application bridge

Exact source line ranges, accepted byte-identical counterpart and original code commit are recorded. Static code is not fresh execution.. Exact locators in structured key.

W-C5: QPU encoding and contracts

Exact source line ranges, accepted byte-identical counterpart and original code commit are recorded. Static code is not fresh execution.. Exact locators in structured key.

W-C6: Output/save boundary

Exact source line ranges, accepted byte-identical counterpart and original code commit are recorded. Static code is not fresh execution.. Exact locators in structured key.

WS-appendix: EWS appendix

explanatory_source. docs/showcase/ephemeral_workspace_v01/technical_appendix_v01.md.

WS-claims: EWS claims

claim_navigation. docs/showcase/ephemeral_workspace_v01/claim_evidence_matrix_v01.json.

WS-index: EWS index

historical_source_index. docs/showcase/ephemeral_workspace_v01/source_index_v01.json.

WS-P_controlled_0: EWS controlled_0

PRINCIPAL_CONTROLLED_LATER_DERIVATIVE. docs/showcase/ephemeral_workspace_v01/presentation_evidence/semantic_boundary_io_projection_v01.json.

WS-P_controlled_2: EWS controlled_2

PRINCIPAL_CONTROLLED_LATER_DERIVATIVE. docs/showcase/ephemeral_workspace_v01/presentation_evidence/semantic_boundary_io_projection_v01.json.

WS-P_topology: EWS topology

PRINCIPAL_EWS3R2_VIA_EWS4R_SAFE_DERIVATIVE. docs/showcase/ephemeral_workspace_v01/public_safe_package/runtime_execution_topology_safe_v01.json.

WS-render_code: EWS render_code

render, lines 100-125. hedgehog/domains/ephemeral_workspace/media_v01.py.

WS-session_code: EWS session_code

execute_effect/save_candidate/approve/close, lines 241-366.
hedgehog/domains/ephemeral_workspace/session_runtime_v01.py.

WS-P_sidecar: EWS sidecar

PRINCIPAL_EWS3R2_EXACT_SELECTED_SAVED_CONTENT. Exact locators in structured key.

WS-P_complete: EWS complete

PRINCIPAL_EWS3R2_NAMED_ARCHIVE_MEMBER. Exact locators in structured key.

WS-r2_report: EWS r2_report

external_archive_member. Exact locators in structured key.

WS-P_temporal: EWS temporal

PRINCIPAL_EWS3R2_NAMED_ARCHIVE_MEMBER. Exact locators in structured key.

WS-P_consumption: EWS consumption

PRINCIPAL_EWS3R2_NAMED_ARCHIVE_MEMBER. Exact locators in structured key.

WS-P_events: EWS events

PRINCIPAL_EWS3R2_VIA_EWS4R_SAFE_DERIVATIVE.

docs/showcase/ephemeral_workspace_v01/public_safe_package/session_event_summary_v01.json.

WS-continuation_code: EWS continuation_code

consume_delta, lines 305-358. hedgehog/domains/ephemeral_workspace/media_continuation_v01.py.

WS-P_save_packet: EWS save_packet

PRINCIPAL_EWS3R2_VIA_EWS4R_SAFE_DERIVATIVE.

docs/showcase/ephemeral_workspace_v01/public_safe_package/sidecar_write_packet_safe_v01.json.

WS-P_save_receipt: EWS save_receipt

PRINCIPAL_EWS3R2_VIA_EWS4R_SAFE_DERIVATIVE.

docs/showcase/ephemeral_workspace_v01/public_safe_package/sidecar_write_receipt_safe_v01.json.

WS-P_tearardown: EWS tearardown

PRINCIPAL_EWS3R2_VIA_EWS4R_SAFE_DERIVATIVE.

docs/showcase/ephemeral_workspace_v01/public_safe_package/teardown_summary_v01.json.

WS-service_code: EWS service_code

close, lines 195-220. hedgehog/domains/ephemeral_workspace/local_services_v01.py.

WS-P_memory: EWS memory

HISTORICAL_EWS3_MEMORY_LATER_DERIVATIVE. docs/showcase/ephemeral_workspace_v01/presentation_evidence/historical_semantic_memory_summary_v01.json.

WS-P_saved_information: EWS saved_information

HISTORICAL_EWS3_NAMED_ARCHIVE_MEMBER. Exact locators in structured key.

WS-P_measured: EWS measured

HISTORICAL_EWS3_NAMED_ARCHIVE_MEMBER. Exact locators in structured key.

WS-P_live_accounting: EWS live_accounting

HISTORICAL_LIVE_LATER_DERIVATIVE. docs/showcase/ephemeral_workspace_v01/presentation_evidence/semantic_boundary_io_projection_v01.json.

WS-P_controlled_1: EWS controlled_1

PRINCIPAL_CONTROLLED_LATER_DERIVATIVE. docs/showcase/ephemeral_workspace_v01/presentation_evidence/semantic_boundary_io_projection_v01.json.

X-TD-NAV: Cross-case TD-NAV

Resolve Supplier S1, Airline attempt 04 and X1 actual sources; no slide treated as execution proof.

docs/showcase/two_domain_master_v01/hedgehog_os_two_domain_master_showcase_sources_v01.json.

X-X1: Cross-case X1

Distinct source heads, differences, common authority laws, non-claims and audit/replay routes. docs/evidence/two_domain_all_real_sealed_evidence_program_v01/cross_domain_evidence_index_v01.json.

X-S-REPORT: Cross-case S-REPORT

public-safe recorded S1 provider-origin execution report; raw prompts/responses excluded. docs/evidence/two_domain_all_real_sealed_evidence_program_v01/supplier_water_filter/supplier_safe_execution_report_v01.json.

X-S-INDEX: Cross-case S-INDEX

nine-row historical scenario index; controlled probes and live-bound deterministic rows explicitly classified. docs/evidence/two_domain_all_real_sealed_evidence_program_v01/supplier_water_filter/supplier_safe_evidence_index_v01.json.

X-A-REPORT: Cross-case A-REPORT

public-safe historical attempt 04 execution report; separate from full-stack claim-matrix history. docs/evidence/two_domain_all_real_sealed_evidence_program_v01/airline/airline_safe_execution_report_attempt_04_v01.json.

X-A-MATRIX: Cross-case A-MATRIX

Preserved failed/recovery distinction and signature_verified=false; never imported into attempt 04. docs/showcase/airline_all_real_full_stack_v01/claim_evidence_matrix_v01.json.

X-LS-PROJECTION: Cross-case LS-PROJECTION

LS2R2 public safe-derived fields, not raw provider history or whole-history execution seal. docs/showcase/landslide_sentinel_v01/public_safe_package/projection.json.

X-LS-MATRIX: Cross-case LS-MATRIX

explanatory source-bound index; source_commit 2f328be634247be11bc18a3b22a919f393d6ed1d. docs/showcase/landslide_sentinel_v01/claim_evidence_matrix_v01.json.

X-LS-CLAIMS: Cross-case LS-CLAIMS

archive/member/hash routing and claim boundaries; external originals not reopened by worker. docs/showcase/landslide_sentinel_v01/claim_evidence_map_v01.json.

X-LS-SOURCE: Cross-case LS-SOURCE

reviewed implementation-body identities and bounded test-setup distinction, not execution evidence. docs/showcase/landslide_sentinel_v01/source_index_v01.json.

X-ATLAS: Cross-case ATLAS

canonical 68250583-byte INCIDENT_ATLAS_AT5_V01 saved proof; selected subtrees inspected, not a claim of reading all embedded code. docs/showcase/incident_to_proof_atlas_v01/evidence/atlas_package.json.

X-AT-CASES: Cross-case AT-CASES

ATLAS_EXACT_CARD_INDEX_V02; card pointers and causal-scope labels.
docs/showcase/incident_to_proof_atlas_v01/reader/CASE_ROUTES.json.

X-AT-CLAIMS: Cross-case AT-CLAIMS

ATLAS_EDITORIAL_CLAIM_ROUTES_V02; C03/C05/C06/C07/C10-C17 package references, not additional proof. docs/showcase/incident_to_proof_atlas_v01/reader/CLAIM_ROUTES.json.

X-AT-CONSUMERS: Cross-case AT-CONSUMERS

five actual consumer routes and effective sample accounting.
docs/showcase/incident_to_proof_atlas_v01/reader/CONSUMER_ROUTES.json.

X-AT-BINDINGS: Cross-case AT-BINDINGS

static source selection, distinct historical phase hashes and AT6 landed bodies; not an import closure.
docs/showcase/incident_to_proof_atlas_v01/reader/SOURCE_BINDINGS.json.

X-AT-REVIEW: Cross-case AT-REVIEW

saved independent source-and-evidence review of the September 23 AT5 return, no archived code execution or test reruns.
docs/showcase/incident_to_proof_atlas_v01/evidence/review/AT5_INDEPENDENT_REVIEW.json.

X-AT-EXTERNAL: Cross-case AT-EXTERNAL

dated September 23 external-source catalog; worker read local entries only, did not verify web claims.
docs/showcase/incident_to_proof_atlas_v01/presentation/external_sources.json.

F01: Permitted schedule and native offer

Historical G54D1 p7_candidate_f779b790; revoked neighbor p7_candidate_7071661f. Original archive and member identities retained, not the later G6A Football.. Exact locators in structured key.

F02: Source body and requester consumption

Historical G54D1 p7_candidate_f779b790; revoked neighbor p7_candidate_7071661f. Original archive and member identities retained, not the later G6A Football.. Exact locators in structured key.

F03: Revoked-source neighbor

Historical G54D1 p7_candidate_f779b790; revoked neighbor p7_candidate_7071661f. Original archive and member identities retained, not the later G6A Football.. Exact locators in structured key.

F04: Separate booking and history

Historical G54D1 p7_candidate_f779b790; revoked neighbor p7_candidate_7071661f. Original archive and member identities retained, not the later G6A Football.. Exact locators in structured key.

E01: Indices and Source Views

Indices and Source Views. docs/gate6_reference_v01/evidence_index.json;
docs/gate6_reference_v01/source_impact.json.

E02: Exact G3 Numerical Rules

Exact G3 Numerical Rules. hedgehog/outcome_calibration_v01.py.

E03: Supplier Refusal -> History -> Different Review

Supplier Refusal -> History -> Different Review.

docs/showcase/gate3_closure_v01/evidence/g36r/runtime/captured_final/report.json.

E04: G35 Does Not Manufacture Scorable History

G35 Does Not Manufacture Scorable History. docs/gate6_reference_v01/evidence/blobs/ec4ad7b768ca5790a3c261d2797b62a9f4992d314fd31bd2ea5104aad0ceb97e.

E05: Worked G4 Allocation, Exact Recorded Numbers

Worked G4 Allocation, Exact Recorded Numbers. docs/gate6_reference_v01/evidence/blobs/a0d0753edb2e6d235fdc18a14eb184a9d8d7f9c8d77eecaced3c679fb5443db9; docs/gate6_reference_v01/evidence/blobs/cfc0bf8da5e79b2778b3383d6d20e1dff09f02ad213cfc779128089b2488466d; 4 additional source locators in structured key.

E06: Current History Descent

Current History Descent. docs/gate6_reference_v01/evidence/blobs/d7603a6812fc2f14a917b8891e2aceaebfd60a718fd9eab816b702fbefcd825a; docs/gate6_reference_v01/evidence/blobs/1a173e73b908da52b3cbb3a09220a2d5d2f1d2dbd46bc35fa7ef8408a57270d5; 1 additional source locators in structured key.

E07: Choice Is Not Shared Authority

Choice Is Not Shared Authority. docs/gate6_reference_v01/evidence/blobs/a0d0753edb2e6d235fdc18a14eb184a9d8d7f9c8d77eecaced3c679fb5443db9; hedgehog/gate4_strategy_reference_v01.py.

E08: N4-04 Cold/Warm Reuse

N4-04 Cold/Warm Reuse. docs/gate6_reference_v01/evidence/retained_obligations/reuse/witness.json; docs/gate6_reference_v01/evidence/retained_obligations/reuse/checked.json; 2 additional source locators in structured key.

E09: N4-05 Current Policy and Restricted Descent

N4-05 Current Policy and Restricted Descent.

docs/gate6_reference_v01/evidence/retained_obligations/policy/witness.json;

docs/gate6_reference_v01/evidence/retained_obligations/policy/checked.json; 1 additional source locators in structured key.

E10: High Score Does Not Supply Permission

High Score Does Not Supply Permission.

docs/gate6_reference_v01/evidence/retained_obligations/avf/checked.json;

demo/run_gate6_retained_obligations_v01.py.

E11: Nested Work and Memory: What Is Actually Supported

Nested Work and Memory: What Is Actually Supported. hedgehog/kernel/work_composition_v01.py;

hedgehog/capability_memory_binding_v01.py; 1 additional source locators in structured key.

E12: Deferrals and Non-Claims

Deferrals and Non-Claims. docs/gate6_reference_v01/acceptance_matrix.json;
docs/gate4_reference_contract_v01.md; 2 additional source locators in structured key.

RW01: Google OR-Tools. CP-SAT Solver

Integer constraints and explicit solver statuses.. https://developers.google.com/optimization/cp/cp_solver.

RW02: Temporal Technologies. Tasks

Durable execution and recorded results.. <https://docs.temporal.io/tasks>.

RW03: Open Policy Agent. Documentation

Policy decisions and enforcement.. <https://www.openpolicyagent.org/docs>.

RW04: seL4 project. capDL

Kernel objects and capability distributions.. <https://docs.sel4.systems/projects/capdl/>.

RW05: W3C. PROV Overview, Working Group Note, 30 April 2013

Provenance concepts, not truth or authority.. <https://www.w3.org/TR/2013/NOTE-prov-overview-20130430/>.

RW06: Prime Minister of Australia. Press conference, New York; recorded publication date 24 September 2026

External incident motivation only.. <https://www.pm.gov.au/media/press-conference-new-york>.

RW07: European Commission AI Act overview, Application timeline and simplification

Dates attributed to the Commission page. EUR-Lex fresh access returned a JavaScript challenge. No legal classification or conformance claimed..

<https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:02024R1689-20260727>;

<https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai>.

RW08: NIST. AI Risk Management Framework

Voluntary risk-management context.. <https://www.nist.gov/itl/ai-risk-management-framework>.

RW09: ISO/IEC. ISO/IEC 42001:2023 public abstract

Management-system scope; paid full text not inspected.. <https://www.iso.org/standard/42001>.

OR-ATLAS: Incident origins and exact case routes

Context or engineering translation, not fresh runtime evidence. references/current/external_sources.json.

OR-SENTINEL: Research-to-Sentinel translation

Context or engineering translation, not fresh runtime evidence.

references/current/research_context_v01.json.

RW10: Microsoft IFC and FIDES context

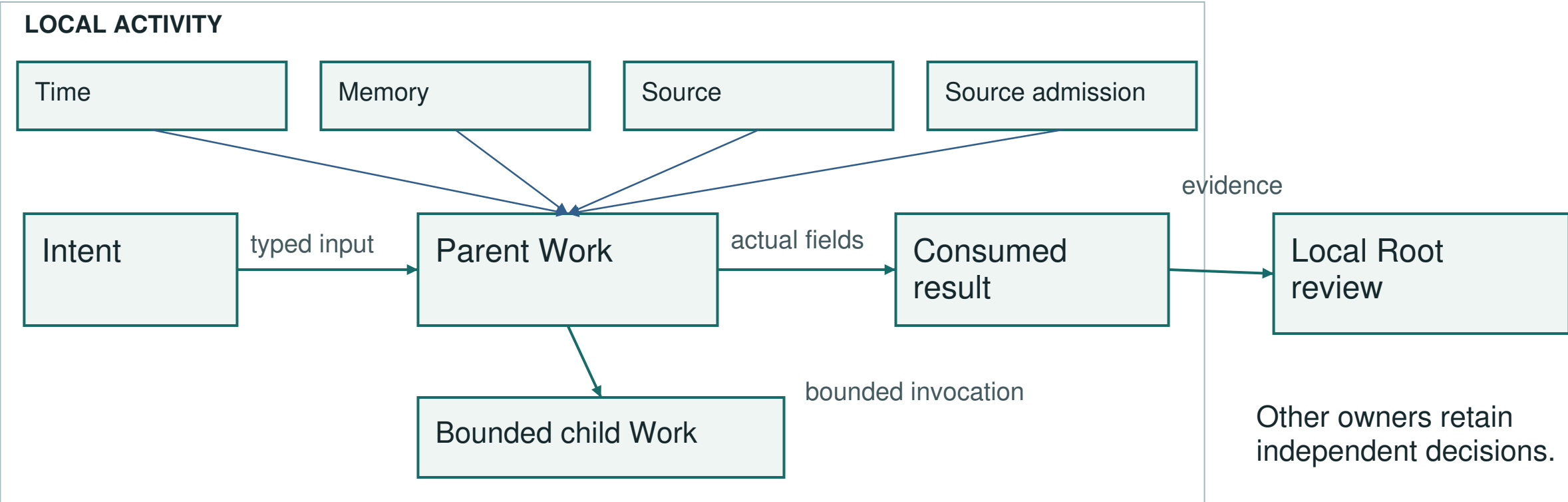
Context or engineering translation, not fresh runtime evidence. references/current/external_context.json.

RW11: Apple manufacturer context

Context or engineering translation, not fresh runtime evidence. references/current/external_context.json.

One activity; several independent concerns

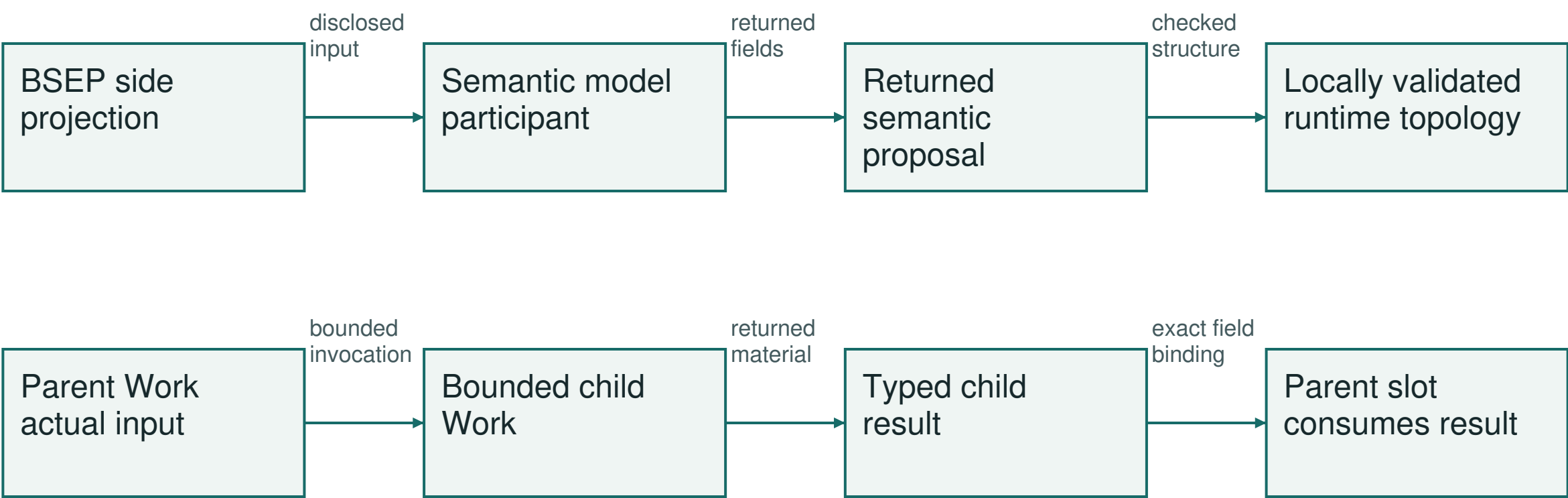
CONCEPTUAL MAP / Relations are not a recorded composite execution or a global owner



Time, memory and source qualify use. Source admission binds implementation. Current effects need their own packet.

Intent becomes bounded, consuming Work

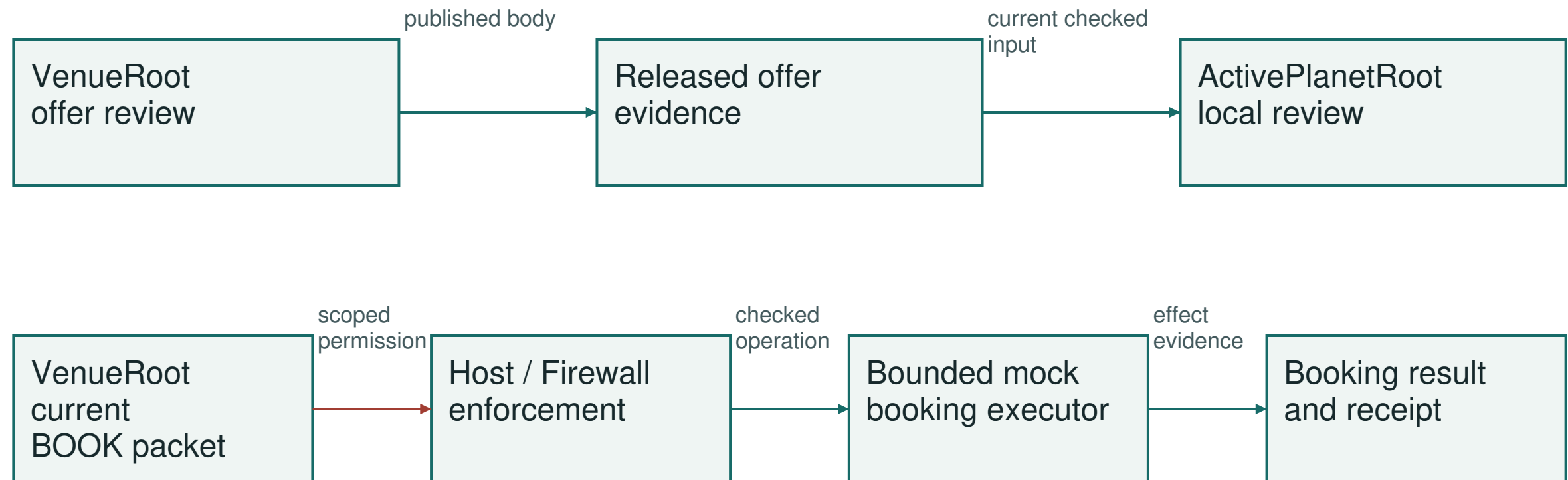
CONCEPTUAL / Local intake and route acceptance precede semantic composition



Root route/result reviews remain distinct. No arbitrary planner or universal child-memory search is demonstrated.

Independent owners retain their own decisions

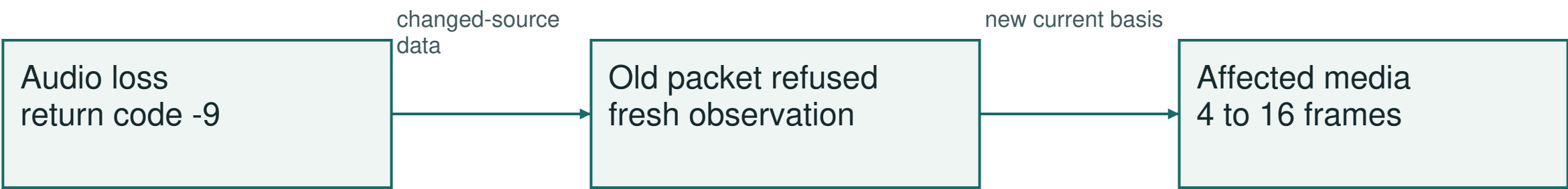
CONCEPTUAL PATTERN + FOOTBALL INSET / VenueRoot and ActivePlanetRoot are separate instances



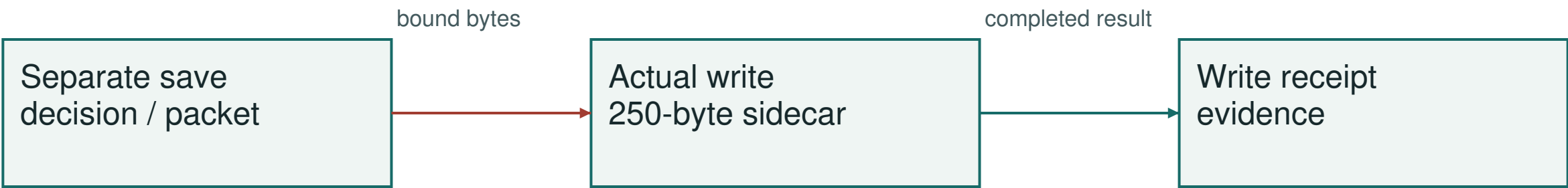
Airline is a separate case with ClientRoot, AirlineRoot and BankRoot. No common business SuperRoot or new action service.

Useful work can outlive a changed resource

RECORDED EWS3R2 PRINCIPAL LANE / Separate clocks below, no receipt-controlled cleanup



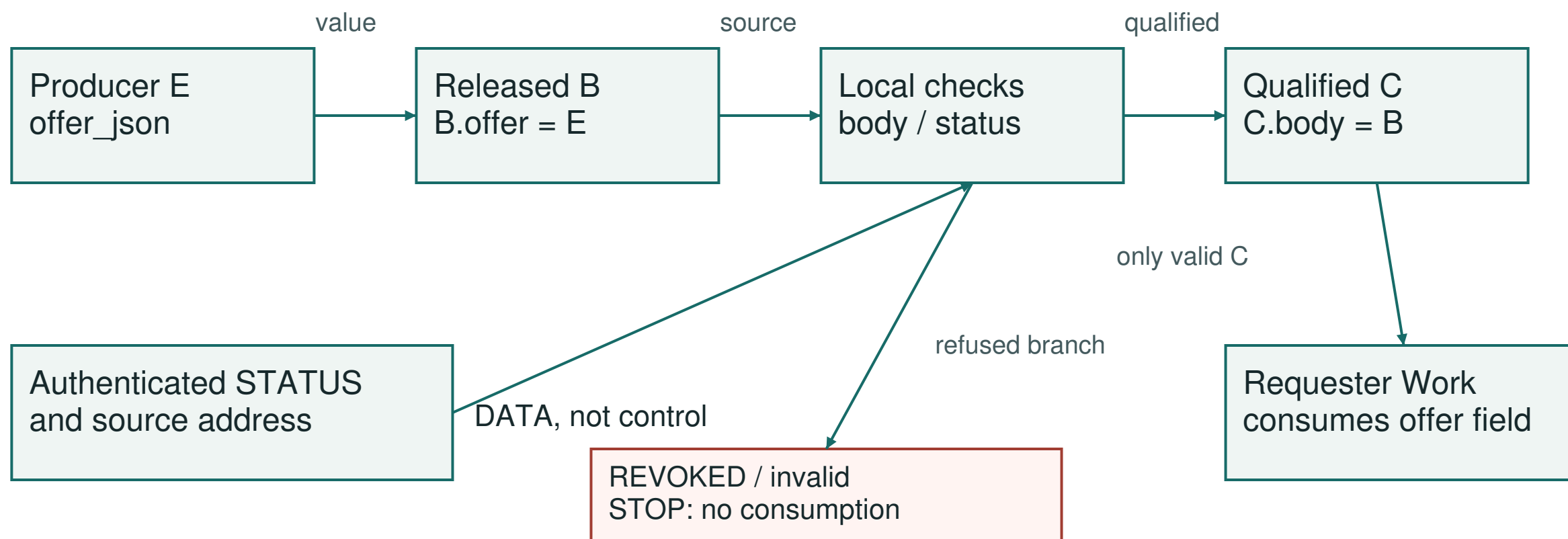
RETAINED PHOTO Same state and preview SHA. Producer count remains 3.



RESOURCE END: owner-lifecycle cleanup. Audio ALREADY_EXITED, no graceful receipt.
HISTORICAL EWS3: separate summary, later SUMMARY_ONLY descent and fresh Root review.
Save precedes recorded closure chronologically. The receipt does not command cleanup or authorize another write.

Qualified context reaches Work; refusal stops

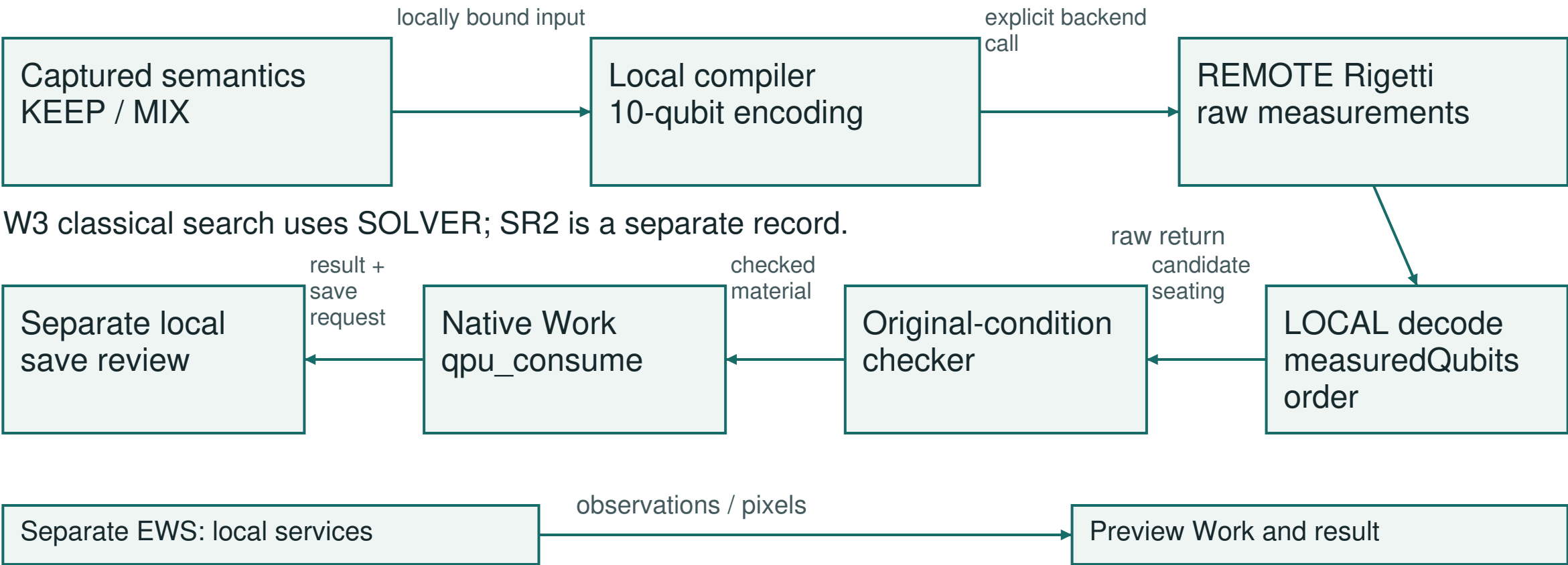
RECORDED FOOTBALL RELATIONS / E is a value inside B; B is material inside C



No arrow leads from refusal to Work. Booking still requires a separate current Root packet and action.

Different computations share checked consumption

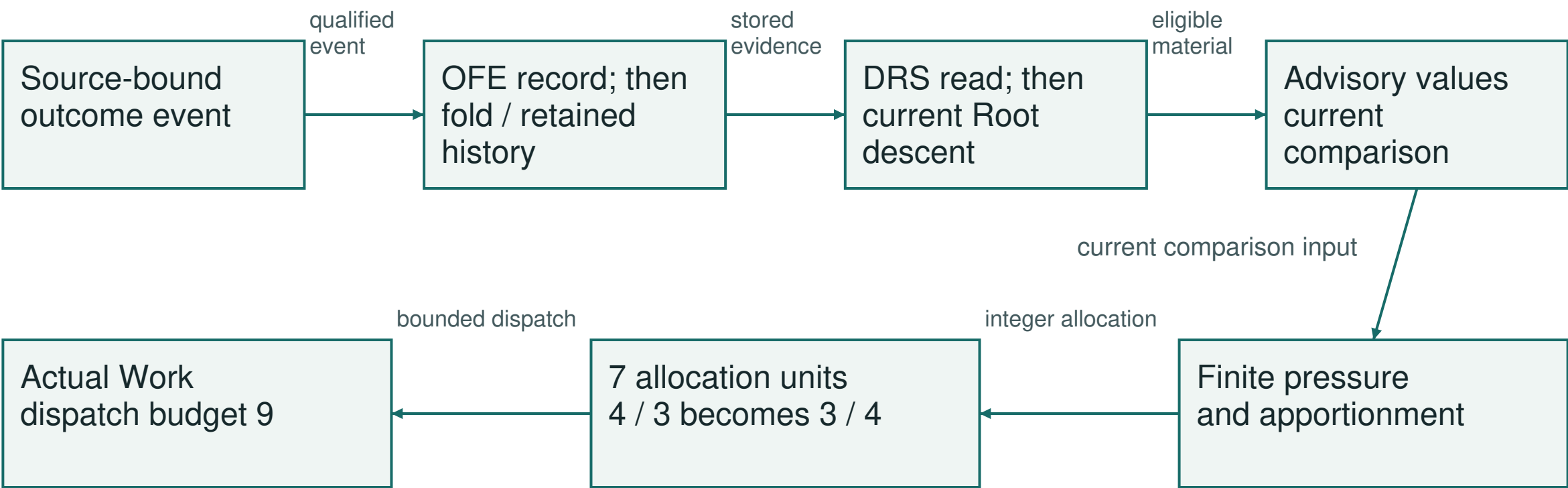
RECORDED W4 REVISION 1 / Captured W3 meaning is input, not permission for a QPU request



36-indicator QUBO differs from ten-qubit up-to-four-local encoding. The EWS inset is a separate local environment.

Recorded experience changes finite expenditure

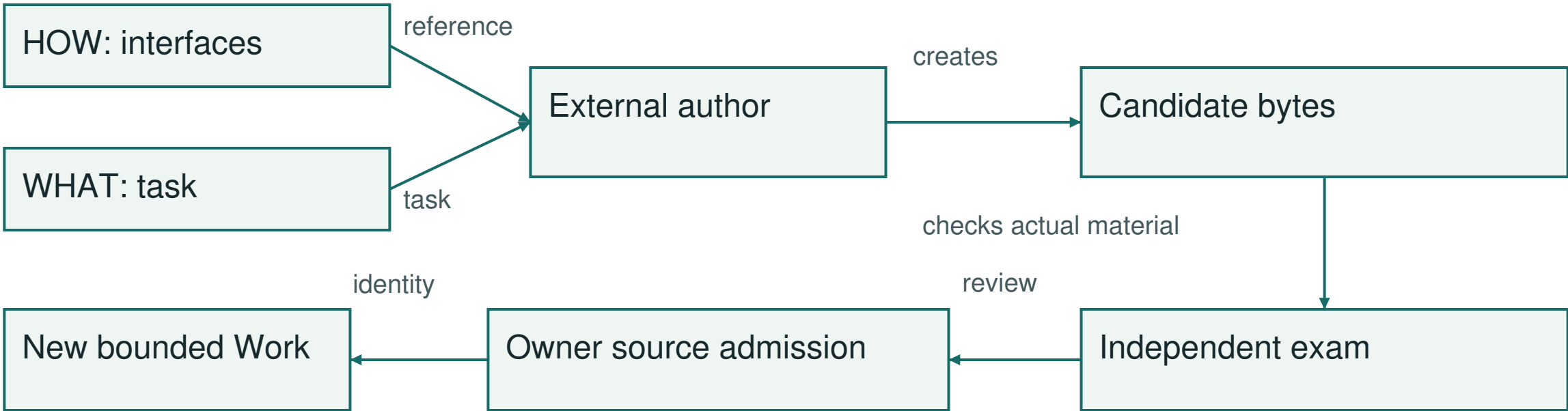
G3 / G4 RECORDED MECHANISMS / Eligibility and Root descent precede current advisory use



Budget 9 = 1 spent + 1 reserved + 7 allocated. N4-04 is a separate 1-to-0 domain-call witness with current reads and reviews.

Creating a component does not grant its authority

CONCEPTUAL / HOW and WHAT are separate; candidate material is not its author



New current inputs and Root decisions remain required. No stored runtime grant.

G54B1/G54D remain INCOMPLETE. G54D1 includes explicitly authorized correction 4; the audit packet is not the kit.